Carpet-Bomb Studios

Testing and exploration of technical specifications with Hardware & Software

Device Testing Portfolio

This document was created for the Carpet-Bomb Studios company in the aid of developing the video game designs for different devices and platforms.

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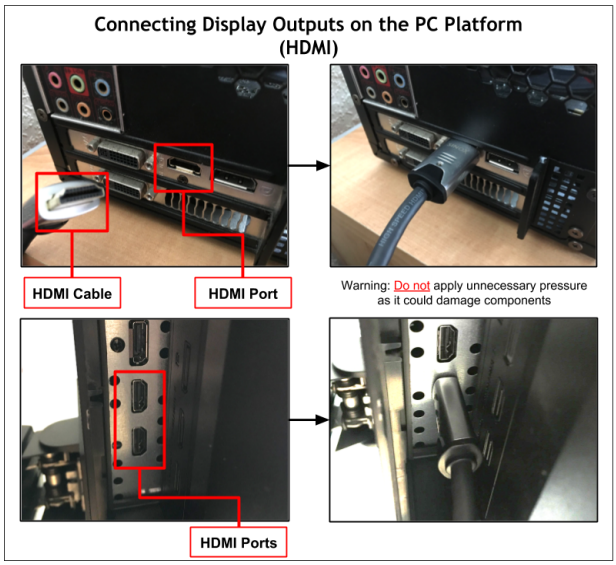
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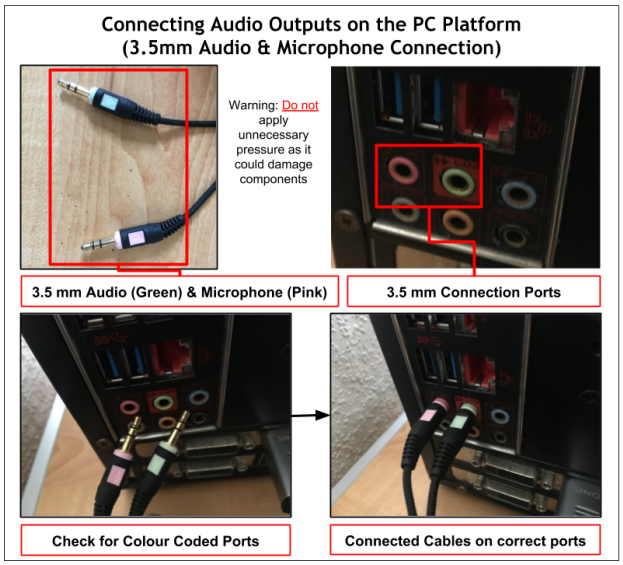
Connecting & Configuring of Platforms & Devices

PC

Connecting of Platform / Device

Relevant Outputs



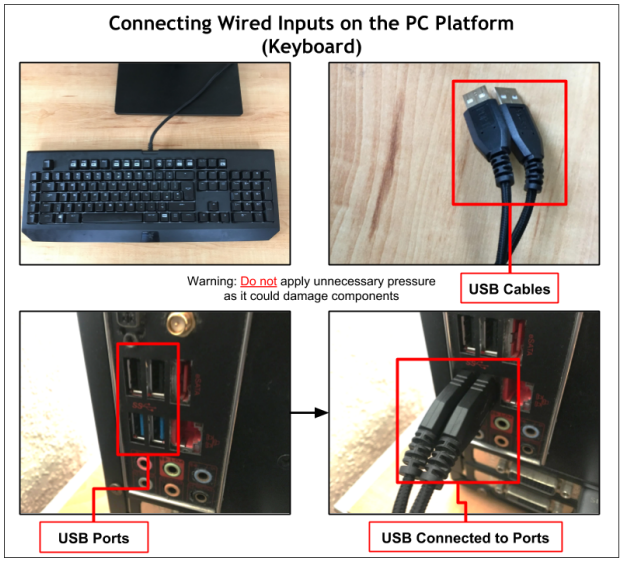


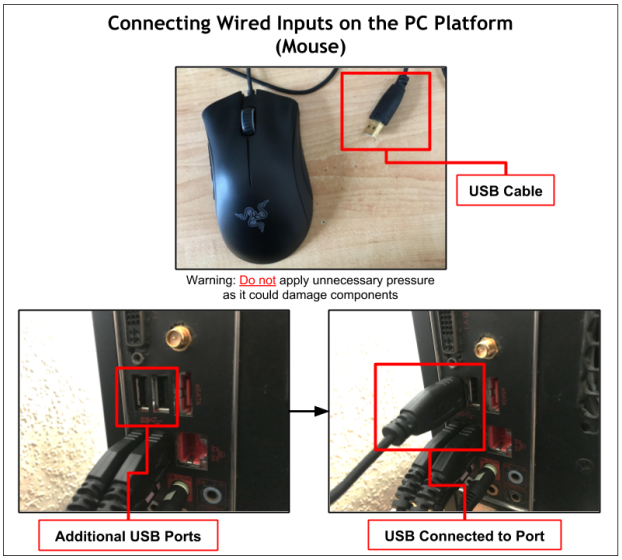
The left image illustrates the steps required to connect a display output from the PC. Showing the ports on both the PC and the Monitor and what type of cable is needed for the said ports, through the means of a HDMI cable. The right image illustrates the steps required to connect an audio output from a pair of headphones to the PC, along with a microphone connection. In this case, it is through two 3.5mm jacks with one as an Audio connection and the other as a Microphone connection.

To elaborate, the image on the left is a basic step by step guide that is annotated on how to install the appropriate cabling in order to secure a direct display output, with HDMI cabling and ports. While the guide on the right is showing the different types headphone jacks (or connections) that are commonly found, especially with headphones that have an integrated microphone. Indicating which ports they are supposed to be inserted on, for adequate use.

Basically, these guides help on how to install the appropriate cabling for the PC outputs in order to secure a direct display, audial and microphone output. With a HDMI cable and x2 3.5mm Audio Jacks, with one as an audio connection and the other as a microphone connection.

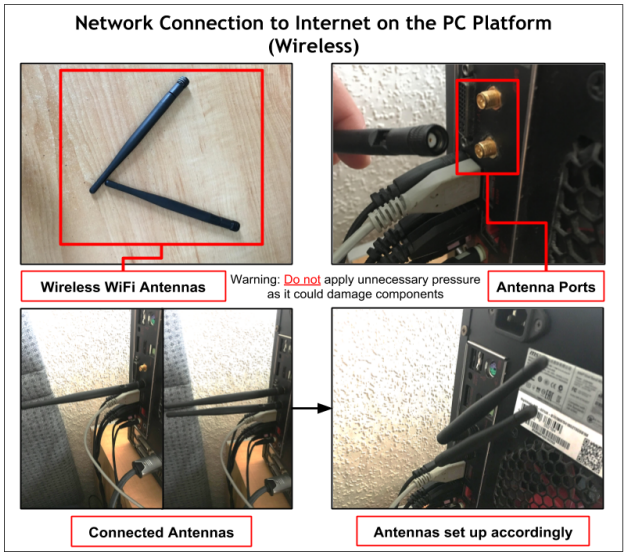
Relevant Inputs

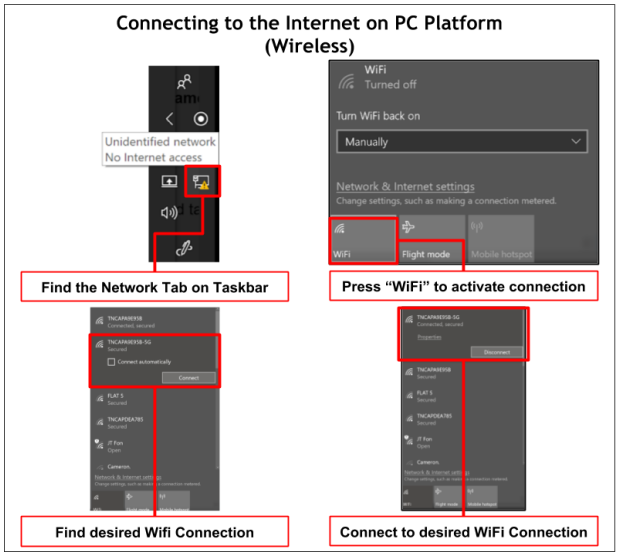




The image on the left is illustrating the keyboard, with the USB cables that come along with the said keyboard. Also illustrating the ports within the PC that are compatible with the USB cables provided from the keyboard. While the right image illustrates a basic mouse alongside its USB cabling. Similar to the keyboard, it will be connected to the USB ports found in the rear of the PC. Essentially indicating additional USB ports found within the PC that give the mouse the ability to be connected.

Internet/Network Connection





The image on the left illustrates the steps required to install Wifi antennas to a PC platform. While the guide on the right illustrates the steps required to set up a Wi-Fi connection via the

antennas to a router providing the connection to the Wireless access point, leading to the connection of the internet.

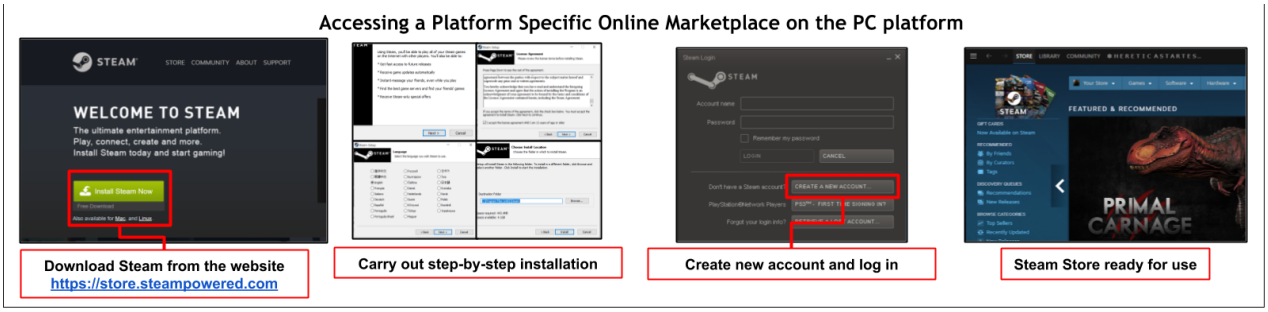
To elaborate, The illustration on the left is a step by step guide on how to properly install antennas into the correct slots, and setting them up accordingly to achieve the maximum potential Wifi strength within the user's home or office.

While the guide on the right illustrates how to activate a wireless connection through the wireless antennas, thus meaning that the antennas must be installed initially. Essentially, showing how to choose and connect to the correct Wifi "hotspot" or router.

Basically, it illustrates the user how to install antennas for the use of Wifi, and how to connect to the Wifi after the installation of the said antennas.

Configuration of Platform / Device

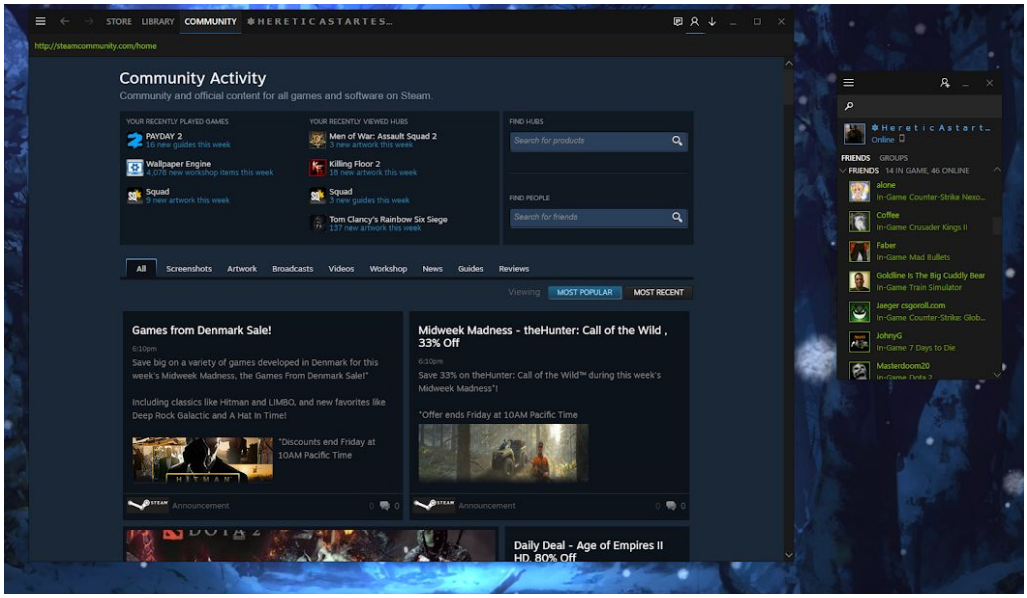
Accessing a Platform Specific Online Marketplace



The illustration above shows how to download the Steam service through Valve's (Creators of Steam) own website, showing the steps necessary to instal Steam without any issues or errors. To elaborate, this shows the quick and user friendly method in which Steam is installed, thanks to Valve's reputable user friendliness when it comes to their services.

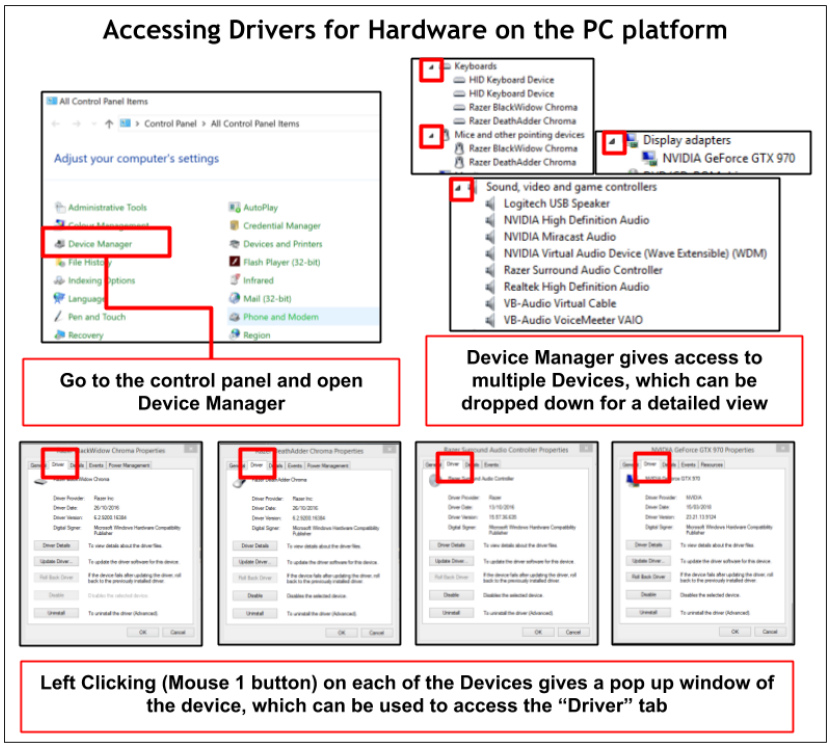
Hence why the step by step guide is so simple in layout. Basically, it's just a simple guide which would be used as a reference point to make sure that the user is installing the correct services with the correct intent.

Accessing a Platform Specific Online Gaming Community



Additionally, Steam does not only offer virtual retail services for video games. Steam also gives the user also the ability of adding Friends and joining groups through the Steam Community. Basically, there is a wide community on the Steam platform where there are millions of active users that are openly available to talk to, play video games with and even make friends with.

Accessing Drivers for Hardware (Graphics, Sound, input and output control mechanisms)



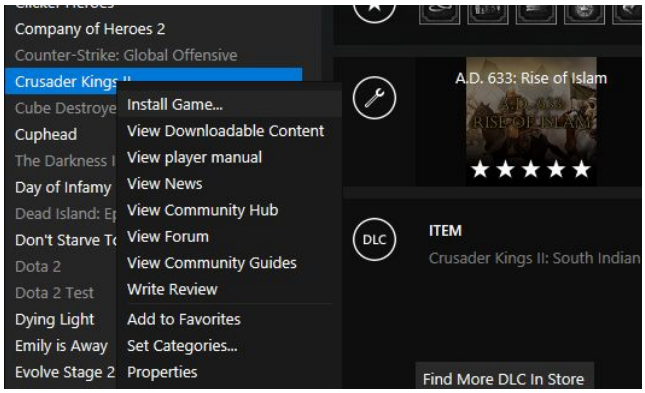
With the image above, it illustrates how to access the different drivers available on the PC platform. To elaborate, it is a step by step guide educating the user on how to access the

different drivers of each hardware device. Showcasing the Graphics, Sound, Keyboard and Mouse Drivers currently installed within the computer.

Basically, this could be of use to the user if they are intending to upgrade their driver, install a new driver or are curious of what driver they currently have and is considering an upgrade.

Installation of a Game

Step 1 - Choosing the Video Game



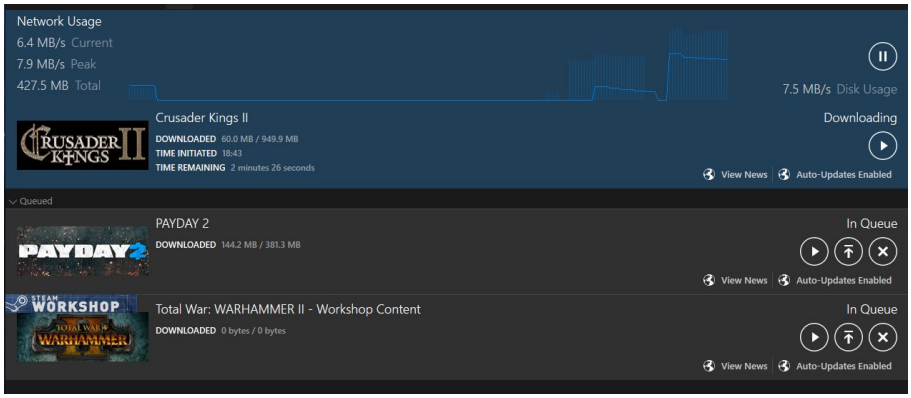
Firstly, choose the desired video game for installment. For this example, I will be installing “Crusader Kings II” to show the simple process of installing a video game on Steam. Which would then lead onto the next step.

Step 2 - Pre-Installation Process



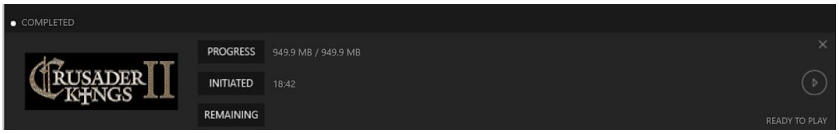
The Pre-installation process is Steam essentially giving the user the ability of choosing which disk they want the game to install in. Which then they would have to agree to the terms and conditions and Steam will start allocating disk space for the game, initiating the installing process automatically, in which it will download until the process is complete.

Step 3 - Downloading the Game



While the game is in the process of downloading, the user can always check the progress of the download by seeing how much data has already been downloaded, how large the full game is in comparison that what has already been downloaded, the estimated time remaining, the internet download speed and the disk install speed.

Step 4 - Installed and Ready for use



When the game is finished installing, Steam will give the player a notification that the game has finished and the game will be ready to play without any further action needed.

Applying / Checking Graphics Settings Appropriate for Hardware Specification

In the PC platform, there are multiple ways in which the graphics setting within a video game can be set for the appropriate hardware specification. Due to the fact that PC has such customisable hardware, it is a vital requirement that video game developers have to include a wide variety of graphical options in order to ader to the vast amount of different hardware capabilities.

The first method is manually changing the graphics within the game to the users desire. This method tends to be very unreliable, especially to a user who isn't experienced with graphical knowledge or knowledge of their hardware capabilities. Usually leading the game to be too demanding for the hardware, causing it to perform poorly.

However, it could also be the contrary, whether intentional or not. It is very common for players to reduce the graphics settings in order to get as much performance from the game as possible. Especially in competitive video games since they would consider extra performance as a competitive advantage.



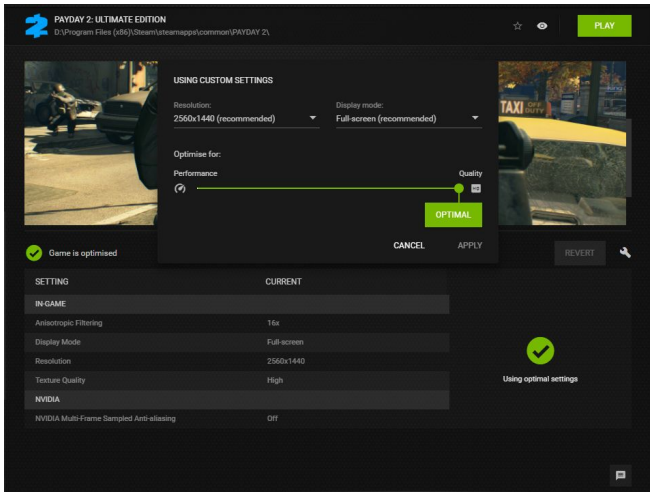
The second method is through research of the recommended and minimum specifications in comparison to what the user has. This is a somewhat desirable way of checking for a game before purchase, to see whether the user can run the game properly in the first place, or if they want to see how their PC might perform on a game they already own.

However, this information can sometimes be unreliable since they can sometimes be very inaccurate or even misleading due to the lack of information on the specific hardware specifications needed.



The last and easiest method, is through Nvidia's GeForce Experience, which is only available for Nvidia GeForce graphics cards. This method of game optimisation is quick and easy, which is recommended for beginners who want an optimal balance between performance and desirable visuals within a game.

Additionally, it has a bar in which the user can manipulate so they can prioritise whether they want more performance or more visual enhancements.

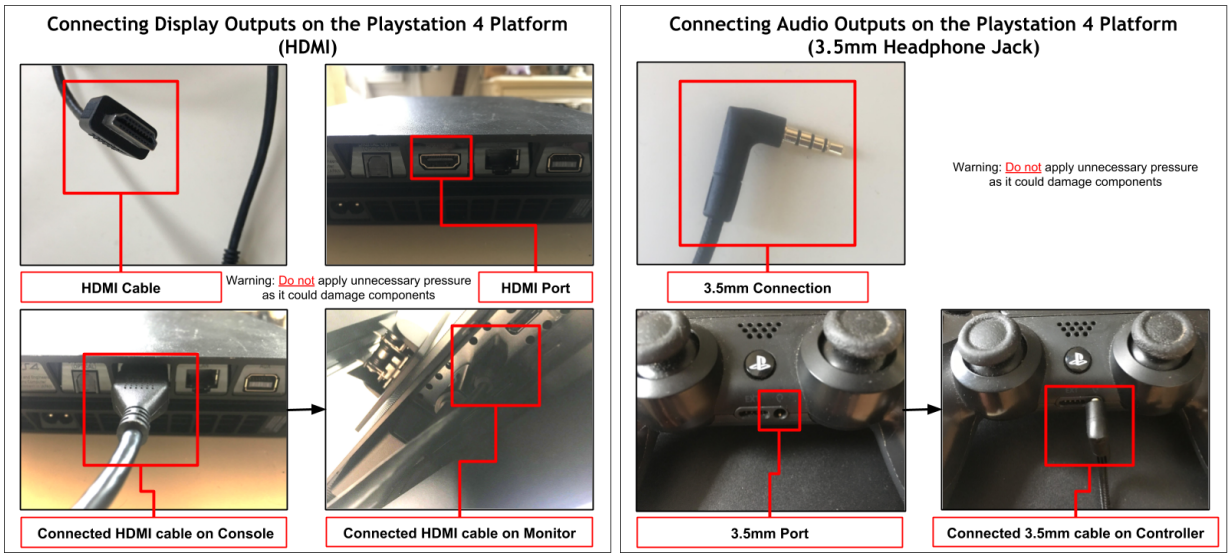


In conclusion, there are multiple ways in which a user can change and apply the graphical settings within a video game. It is completely within the respect on the user on how they desire to manipulate the graphical setting of a video game. Whether it is changing for visual enhancements or better performance.

Console

Connecting of Platform / Device

Relevant Outputs



The images above show the installation process of the relevant outputs on a Playstation 4 platform. The left illustrates the connection of a HDMI port to a Playstation 4 to a Monitor. While the image on the right illustrates the connection of a headset within a Dualshock 4 controller.

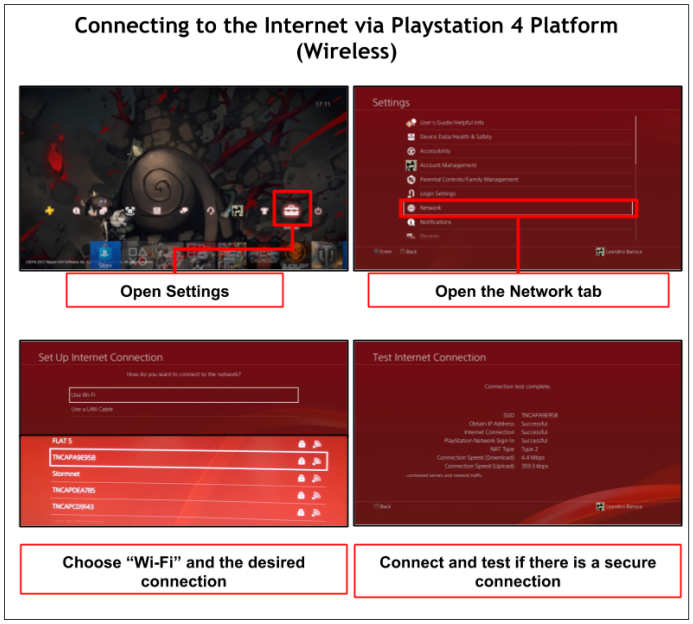
To elaborate, the image on the left is a basic step by step annotated guide on how to install the appropriate cabling in order to secure a direct display output, with HDMI cabling and ports. While the guide on the right is showing the connection for a headphone jack (or connection). Furthermore, since the PlayStation 4 doesn't have a 3.5mm headphone jack within the actual machine, the only connection to an audio output is through the Dualshock 4 controller.

Relevant Inputs



The image above shows the simple process how to connect the Dualshock 4 controller to the Playstation 4. Basically, there is no need on elaboration since the controller only requires the user input for the Playstation button found in the middle of the controller, in which it would give visual and audial feedback through the lights and sounds coming from the Playstation 4 system.

Internet/Network Connection



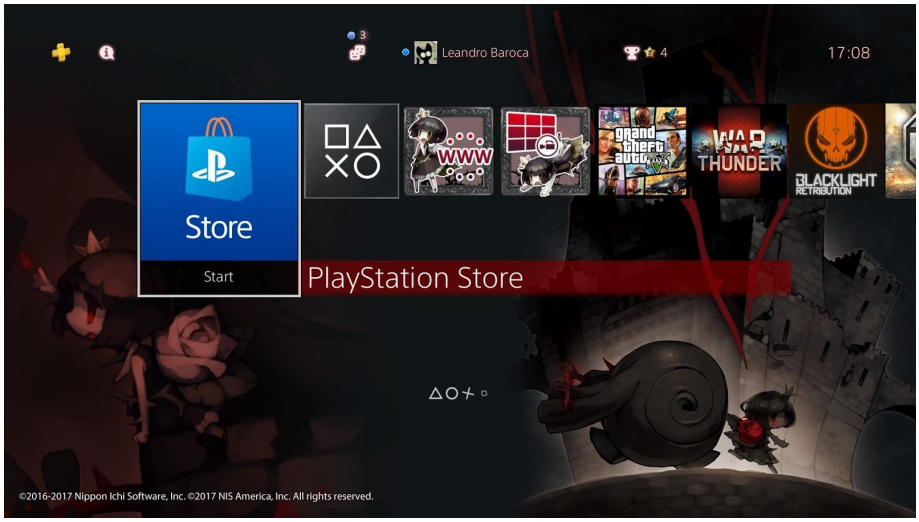
The image above, shows how to set up a wireless internet connection via the PlayStation 4 system.

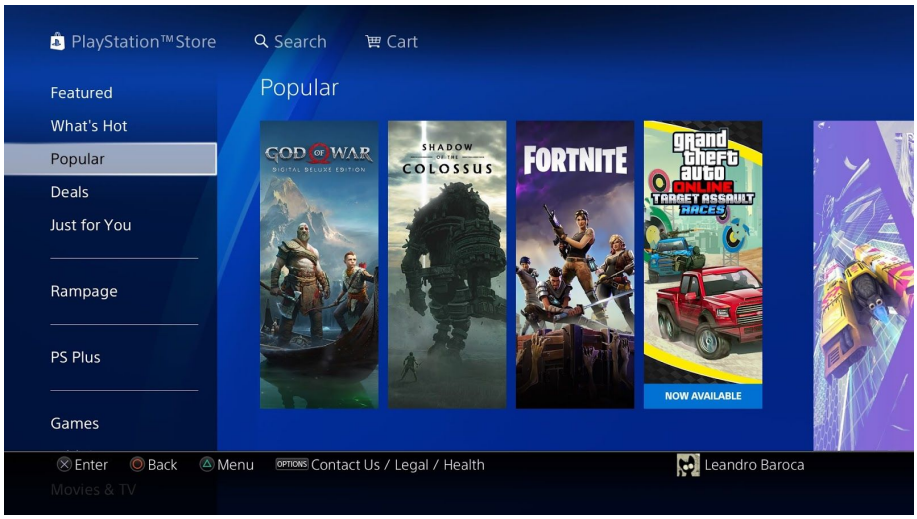
To elaborate, the user would have to navigate through the main menu until they find the setting buttons. In which they would access the settings and then navigate to the network tab. Which they will then select “Set up Internet Connection”, which will then take them through a setup wizard asking them if they desire a Wifi connection or wired connection.

Where the user would choose their desired one. After that they will choose their router or wireless access point and then get taken to a Testing screen to test their internet connection. Essentially, this is a simple step by step guide on how to navigate through the PlayStation menu on how to set up an internet connection.

Configuration of Platform / Device

Accessing a platform specific online Marketplace for Games

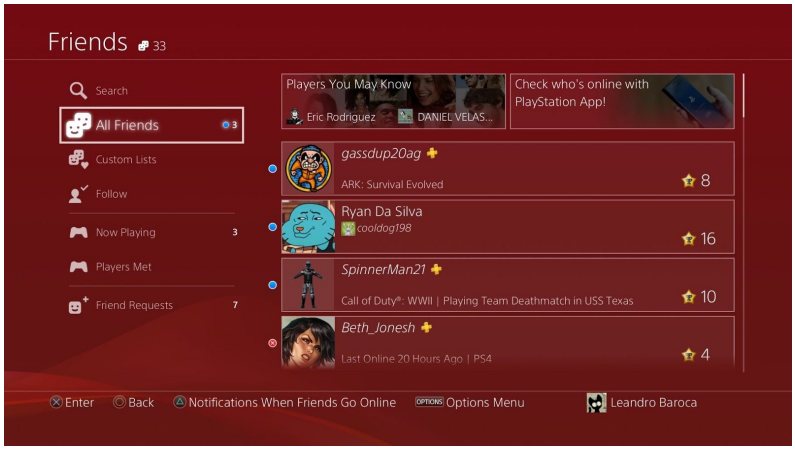




The images above shows the marketplace which is available for the user on the PlayStation menu. To elaborate, this is a marketplace in which the user is able to purchase video games digitally, rather than purchasing a physical copy from a retail store. Essentially, this marketplace enables the user to purchase games much quicker and easier, without the need of installation through a disk. Instead it is through an internet connection.

Accessing a platform specific online gaming community for multiplayer





The images above showcase the gaming community available on the PlayStation 4. Similar that to the Steam community, the Playstation 4 online community also gives the user also the ability of adding Friends and joining groups through the Playstation community.

Basically, there is also a wide community on the Playstation platform where there are millions of active users that are openly available to talk to, play video games with and also make friends with.

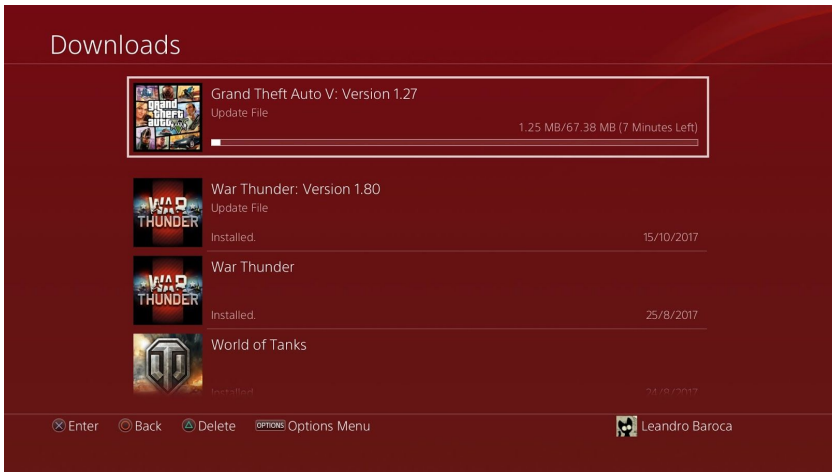
Accessing Drivers for Hardware

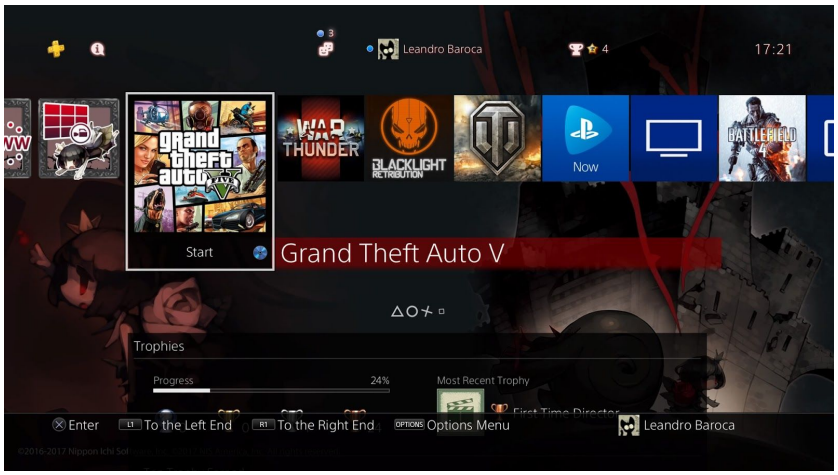




The images above is a software update of a PlayStation 4, which I couldn't take additional images of due to Private information reasons. This update counts as accessing drivers due to the fact the system is upgrading the drivers required to keep the system running with the hardware allocated to the console.

Installation of a Game





The image above is the installation of the Grand Theft Auto 5 video game, which required the installation of the optical disc in order for the game to install. To elaborate, the installation

process of a video game within the PlayStation 4 system is very simple due to the fact it is all completely automated.

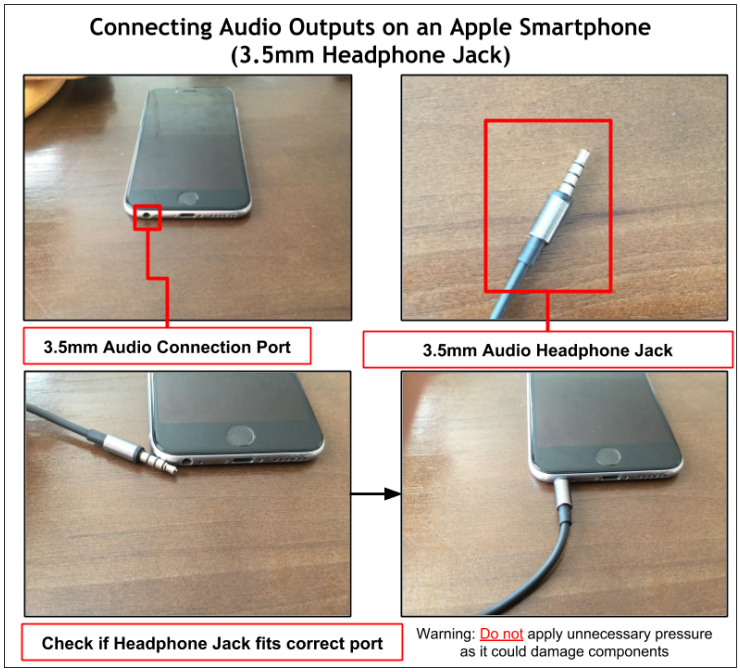
Firstly, the user would need to insert the video game disk, in which it would install the game from the disk and installing any necessary updates, which would bring up the initial screen automatically. After the installation is complete, the player is able to navigate towards the game and launch it with little to no effort needed.

Basically, no step-by-step guide is needed for the installation of a game within the Playstation 4 system due to the vast user friendliness of the system, basically carrying out all the necessary tasks automatically.

Smartphone / Tablet

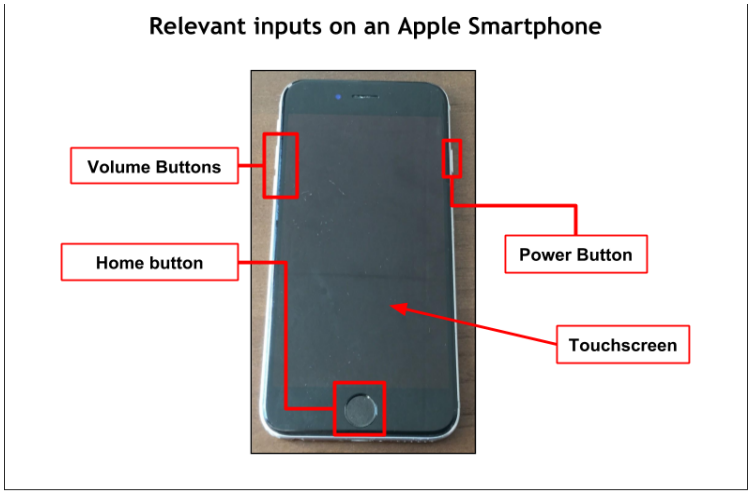
Connecting of Device

Relevant Outputs



The image above shows the connection of an Audio output into an iPhone. To elaborate, this is the only output from a smartphone which can be used by an external cable, which is a 3.5mm headphone jack. This is due to the fact the phone already has most of it's necessary features inbuilt and has no need for further wired equipment. Essentially, this is a simple illustration of what type of headphone jack goes into the iPhone for headphone/earbud use.

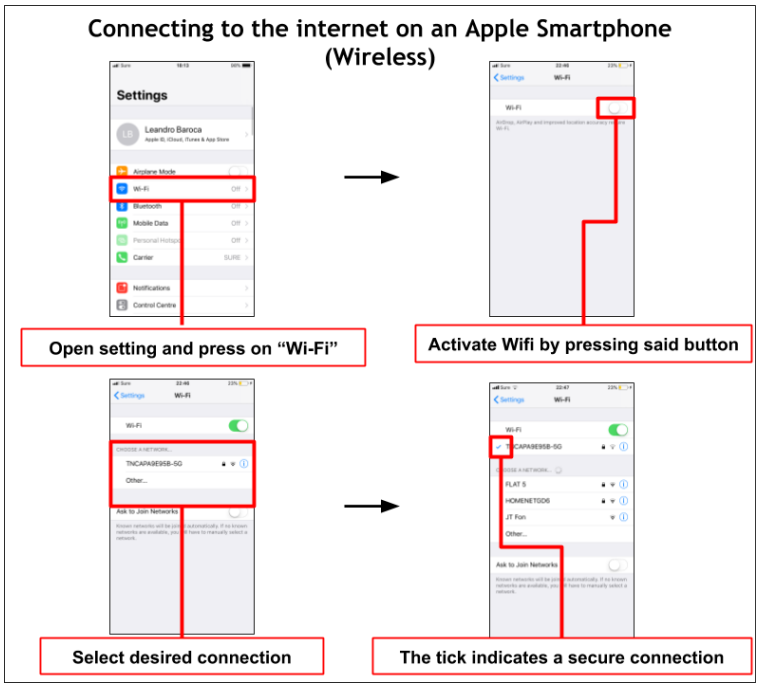
Relevant Inputs



The image above shows the relevant user inputs available within the iPhone. To elaborate, the touch screen function on a smartphone is the most versatile form of user input available for the specific device. The other inputs, such as the power button, home button and volume buttons have specialised and very specific uses that are necessary in the use of the smartphone.

Essentially, the only relevant input available on these devices are already installed in the manufacturing process, and no further input is needed due to the simplicity and sophistication of smartphones.

Internet/Network Connection



The image above is a simple step-by-step process on how to connect to a wireless internet connection over an iPhone. To elaborate, the user would need to go into the setting, which is usually found in the home menu, to then press on the wifi button found in settings.

Afterwards, the user would activate the Wifi connection by pressing the “switch” or button found near the top right of the screen. After that is complete, the user would then choose the desired wireless access point and wait until it gives the verification tick that it has in fact been connected.

Configuration of Device

Accessing a platform specific online Marketplace for Games



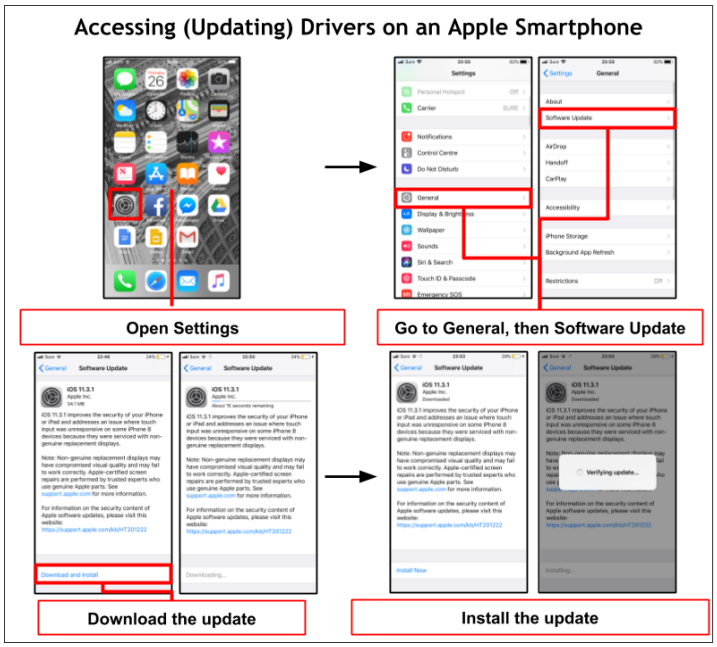
The image above showcases the online marketplace which has the ability to purchase and download games within the Apple smartphone device. Essentially, this is a simple process in which the user must only press on the app store and navigate through the simple yet effective user interface.

Accessing a platform specific online gaming community for multiplayer



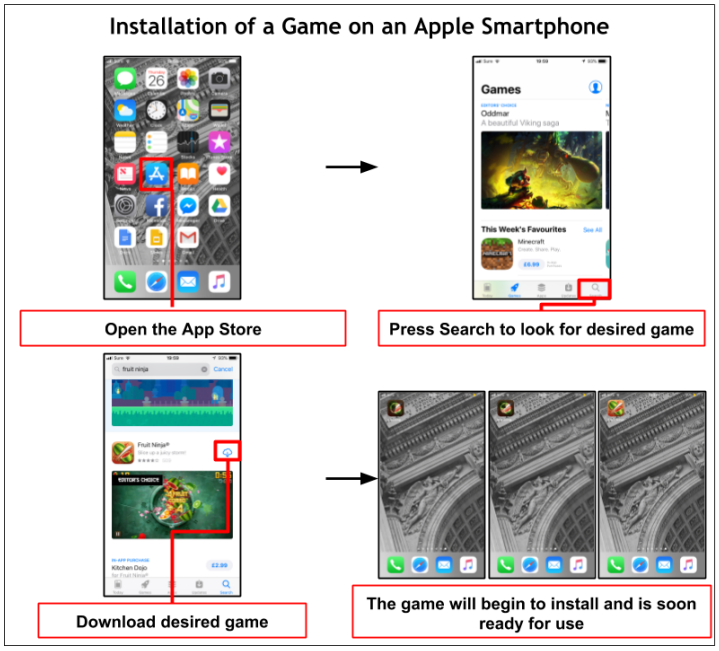
The image above shows the process needed to access a platform specific to an online gaming community, To elaborate, this online community is the Game Center, which is essentially friends from the user's contact list which have been added as "friends" on this community. Which the user will then be able to challenge in different types of video games. Essentially, it is a small community hub in which friends can challenge each other with their favourite games.

Accessing Drivers for Hardware



The image above is how to update the drivers available within the iOS operating system.
To elaborate, drivers in iOS aren't open to the public, meaning that their source code is closed. Therefore, it means that the only way which the user could interact with the drivers of the operating system is through the means of downloading and installing the updates given by Apple.

Installation of a Game



The image above is illustrating the simple process of installing a video game through the app store. To elaborate, the user would have to access the app store which is usually found in the home menu.

Then, the user will have the ability to search for a desired game throughout the search bar found at the bottom right of the screen. After searching for a desired game, they would then click the download button and the application would start downloading automatically and install autonomously. Afterwards, it would then be ready to play for the user.

Hardware Technologies for Configured Platforms & Devices

PC

HCI

Human Computer Interaction, or HCI, refers to the methods that people use to interact with computers. To elaborate, it is how devices are designed and created to allow for humans to interact with machines. Basically, HCI involves interface devices and graphical user interfaces.

Interaction Models

An interface device is a piece of hardware which allows interaction between a person and a computer. Forms of interface device include keyboard and mouse for computers, touchscreens for mobile devices and controllers for consoles.

These interaction models differ greatly depending on what platform or device the user is currently using. Whether it is a PC, Console or even a Smartphone or Tablet, there is always a different interaction model which has been designed to be used by humans for its intended purpose.

For example, the PC would use a keyboard and mouse as it's main form of the user interacting with the software, backed up by the hardware. Giving them full access to the machine and carrying out their intended purpose, which may be video games.

Overall, in the terms of video games, the mainstream HCI devices are (obviously) video game controllers, as well as the keyboard and mouse. However, with the wide accessibility and affordability of the interaction models, there's other reason which led them to be so popular.

Ergonomics

Firstly, is Ergonomic Design. Which is essentially the science of designing a product which has maximum efficiency and productivity. This involves how a product is shaped, what it is made, how much is weights and how overall comfortable it is for the user. If a product isn't ergonomically designed it may cause stress, fatigue or even injury on the user.

With PC gaming, the mouse and keyboard are bulkier and less ergonomic than other forms of gaming controllers. However, it doesn't make it obsolete for gaming. As the individual keys and the ability to move the mouse in any direction makes it much more customisable and the player can adheer the control mechanism as they desire.

For example, the keyboard and mouse was designed to navigate through the software of a PC, by moving the cursor around with the mouse and selecting different icons with the mouse button. While the keyboard was used to type in input in the form of text, commands and hotkeys.

Overall, it goes without a doubt that gaming is more of an afterthought for PC gaming, which had to be adapted to fit the HCI of gaming. As opposed to the gaming consoles, whose HCI has been adapted to fit the games.

Button Configurations

Additionally, Button Configurations refers to the layout of buttons on a device and what they are programmed to do. In some games it is possible to change the button configuration to make it easier for the player to play the game. For instance, poor button configuration would be apparent is if the player needs to press a specific button combination, but it is not possible due to the fact they cannot reach one of the buttons. Hence why button configurations are vital.

For example, the button configuration for keyboards are less efficient when considering video gaming.For instance, most video games played on the computer, the most common keys used for movement are W, A, S, and D. These letters act as the directional pad equivalent to a controller, with the movement of turning and looking is done using the mouse's movement.

This means that, like the controllers, two hands are also needed for the computer but is using a different hand position. The reason that this HCI is less than optimal for the keyboard is because of the position of WASD. While the ASD are all aligned, the W is not aligned with the S key, which would make for a more comfortable position.

This is due to the way the keys are placed for typing, further reiterating how the gaming HCI for computers is an afterthought. Even though gaming on a PC seems to be favourable, the HCI however has different uses when compared to other gaming devices. This is due to the vast amount of keys, that allows for shortcuts and specialised commands, which a controller couldn't replicate.

There are even specific keyboards which include additional buttons, usually located at the top of the keyboard, so that users may equip specific functions to them within games. While also not affecting the intended interaction of a keyboard.

Overall, even though the keyboard and mouse wasn’t initially made for gaming. The ergonomics of the different keys and other commands could be manipulated by the user to fit their needs, making it more customisable than controllers from a console which is completely restricted on it’s button layout.

Inputs & Control Mechanisms

Wired

While wired technology seems obsolete compared to the convenient and perfected wireless technology in the modern day, wired technology can still prove a significant advantage over wireless for video games. For instance, this is due to the fact that wired technology tends to be slightly more responsive than wireless, however that is becoming less apparent with the perfection of wireless technology.

Overall, wired technology seems to be the pinnacle of reliability and efficiency for PC gaming as a method of input for the control mechanisms. However, it is apparent that wireless technology is catching up and being perfected as technology evolves.

Wireless

Wireless technologies within PC gaming is somewhat prominent, however it is widely criticised in PC gaming due to the disadvantages it has when compared to the wired counterpart. Typically there isn’t much difference between wireless and wired control mechanism with PCs, due to the fact it still functions and a keyboard and mouse. For example, it is disadvantageous

because the batteries need to be changed/charged due to the fact that batteries have a limited life span, especially when compared to wired.

Overall, as mentioned before, wireless technology has the potential to overcome the dominant wired input technology that is so widely adored today. However, wired will still have an advantage due to the fact that wireless could always run out of battery eventually, no matter the battery length.

Touch-Screen

While touch-screen technology does exist within the PC market, it is very uncommon to find. Especially within the video game selling point, since it provides no additional benefit to players and it is therefore seen obsolete. Also due to the fact that touch-screen monitors tend to be of worse quality and performance when compared to traditional monitors, which have been perfected over the decades.

Overall, touchscreen technology has potential of being useful for home PC use, but for gaming it is completely obsolete and there isn't any improvement needed for touch screen interaction for video games. Especially with the introduction of VR.

CPU

The CPU, otherwise known as the Central Processing Unit or processor, is the removable chip found on the motherboard of a computer which controls a heavy amount of the processing, essentially as the "heart" or "brain" of the computer. The circuitry inside a CPU processes the instructions that software and programs send it by interpreting the instructions into movements of data, usually found within the computer's memory.

The way this is done is through four functions. The first function is the "Fetch" function, this is when the CPU first received the instructions or "data" from a program. Each of these instructions has a particular and unique address which is kept track by the program counter within the CPU when accessed.

Secondly, is through the Decode function. Essentially, this is when the language in which the instructions are written in are broken down or translated to an Assembly language. Which it could then be understood, therefore using an assembler to translate the code into binary code.

With this phase, the CPU would then be capable to use the code to execute the program's instructions, which brings it to the third function: Execution. The CPU can execute the code by carrying out a necessary procedure of either using its ALU (Arithmetic Logic Unit) to calculate intense and difficult mathematical functions, move a quantity of data within the memory or even jump to various data addresses within the program itself.

The final phase of CPU processing is Writeback. Which is basically when it writes the output into the computer's memory to create the change in data that the program or user requested.

Type

However, as there are multiple different types of devices, there are also multiple types of CPUs. For instance, the two leading CPU manufacturers, Intel and AMD, have their own types of CPU. However, the most significant difference between the CPU are the amount of cores contained. There are three main CPUs that are popular today (While popular, it doesn't mean it is very good for gaming).

Firstly is Dual Core, which is the oldest variant of CPUs. This CPU type can only handle a limited amount of operations at a time, therefore are dreadful at multitasking. Hence why this CPU is the worst type for gaming. The reason for this is due to the "running speed" of a computer being extremely reliant on the processing power of the CPU. Essentially, the use of a Dual Core CPU would mean there would be noticable decreases in performance when a couple application is open and running.

Secondly is Quad Core. Which is essentially 4 Single Cores in one package, hence the name "Quad". This gives the ability for the CPU to handle different streams of data when compared to its predecessor, meaning that it doesn't have to constantly switch data streams as much. Giving it an outstanding upgrade in multitasking abilities and efficiency. Hence why Quad Core is seen as "faster" than Dual Core, hence making the PC "faster" due to the significant reliance.

However, the disadvantage to Quad Core is that it is more expensive than Dual Core, which is expected, but the price is radically reducing due to the introduction of more modern processors which have more cores. Additionally, While Dual and Quad Core must contain a special code stream called SMT (Simultaneous Multi-threading Technology). Meaning that even older processors such as Single Core is less specialised and specific. However SMT is becoming a standard implementation in modern systems, which makes Single Core completely obsolete.

Lastly is Octa Core, which again is a significant improvement to its predecessor as much as the Dual Core was to the Single Core. Quad Core CPU dramatically increases the performance of a PC, especially with SMT, and allows for an even greater multitasking ability, Although, it doesn't necessarily mean that it is four times "faster".

While Octa Core tends to be more expensive, it is significantly more beneficial to those who require many programs open at once, especially for video games. However, video games are advancing rapidly, meaning that Octa Core is also slowly becoming less powerful. Hence the introduction of 16 Core processors, with Intel even introducing 18 core processors with the i9.

Speed

As mentioned above, the speed of the CPU greatly depends on how many cores it contains, however, the speed in which the CPU is processing is also vital to its overall performance. For instance, the speed of a CPU is determined by how fast the processor is working, which is essentially the amount of cycles the CPU can perform within a second.

The performance of a CPU is typically the speed at which it processes the data, universally known as the clock speed. The clock speed is measured in GHz (Gigahertz). For instance, one gigahertz is one billion calculations in the span of one second. Despite this however, it does not

mean that a 2 GHz processor is twice as fast as a 1 GHz processor. Due to the fact that the efficiency of the CPU must also be taken into consideration, which can vary based on the CPU's architecture.

Which is why Clock speed is also used. Essentially, clock speed is what measuring the speed that the microprocessors executes each instruction is called, and the the faster the clock rate, the faster the CPU is. Therefore, the faster applications would perform for the user, such as video games.

Cache

The CPU contains a form of memory storage which is known as the cache, which is either built in or near the CPU within the motherboard. The cache stores functions that are used regularly, which is what creates a computer’s history and allows the system to reload pages through the back/forward buttons.

With modern CPUs, it is more likely to have more than one cache memory storage for different roles, which work together to process information as fast and efficient as possible. Which is very beneficial for video games, since they are the most demanding programs that is commonly downloaded by the average PC user.

Address/Data bus

A vital component of CPU is known as Address/Data bus. Basically, within each device, a system bus is present. This is what essentially connects the major components of a computer system. It typically has three main elements: The address bus, the control bus and the data bus. Which only the address and data buss will be mentioned due to their importance in video game performance.

Firesly, the address bus is used to transfer data between devices that are identified by the hardware address of a physical memory, which takes the form of binary numbers in order to allow the data bus to access memory storage. Essentially, the CPU uses this address bus to locate physical address to communicate commands.

The address bus is measured by the amount of memory a system can retrieve, systems with 32-bit address bus can transfer 4GB (Gigabytes) of address memory space. Whereas the successor to these systems, which use 64-bit address bus, can transfer 16EB (Exabytes) of address memory space. Which is significantly more efficient. Hence increasing in the capability of running higher demanding games at optimal level, with it being constantly improved with the addition on newer CPUs.

The data bus has a similar function, as it is also their to transfer data, only this time it is within the microprocessor and memory devices. A typical data bus is 32 bits wide, which means that up to 32 bits of data can travel through a data bus every second. Just like address buses, 62-bit paths are becoming more popular, and greatly increase the speed of which the CPU functions.

Therefore increasing the performance of video games, especially when run on a more modern PC.

GPU

The graphics card, also known as a video card, is the hardware responsible for displaying 2D and 3D graphics on a connected screen. This is usually done by taking the binary data from the CPU to turn pixels into a picture on a screen.

Most modern motherboards feature integrated graphics built onto the board which can easily handle the creation of 2D images, but using a seperate graphics card will tend to override the integrated graphics, giving it much better performance for 3D graphics.

Type

To begin with there is PCI graphic cards. These use slots called PCI slots that go onto the motherboard to connect to the computer. Now with the introduction of more modern graphics cards, PCI graphics card is considered completely outdated. Despite this however, there are still many computers which contain a PCI slot in the motherboard which lack newer slots, and as such become necessary to use.

Just like the PCI graphic cards, the AGP cards are named after the slot they fit in. Unlike the PCI however, this graphics card can have 4 processes, the fastest being 8x. Due to its varying processing speeds, in turn making it more adaptable to the computer's capabilities. Making it one of the most widely compatible graphics card.

PCI-Express was once the most advanced graphic card available, these can be accelerated to 16x. In addition, motherboards with a PCI-E slot often has multiple others too, which allows more PCI-E cards to be plugged in combining their processing power, much like a dual core CPU.

In the current era, with the introduction of GTX graphics cards from Nvidia, especially the GTX 1080 ti. All the previous graphics card variants seem almost obsolete when in comparison of Nvidia’s GPUs, which are now considered the flagships of modern graphics processing. This is due to the sheer upgrade that Nvidia brought with their innovative architecture and design.

Speed

Due to its similarities to the CPU, the speed of the GPU is also measured using the same clock speed type of measurement, by using MHz. As mentioned previously, the higher the clock speed, the more efficient it is. Therefore becoming more beneficial it becomes to doing complicated calculations.

Which is vital for video games due to the constant and rapid improvement in the graphical capabilities of newer games. For instance, the performance of older GPUs are often downgraded significantly over a short time, because there is constantly more power on newer graphics cards. With the pinnacle being Nvidia's GTX series.

Due to this, the older GPU's clock speed may result in poor game performance, because the processing power is slower or delayed. Whereas the computer with a larger Cache and better processing speed/clock speed may run the game with better performance.

Resulting in higher frame rates and enhancements in visuals, due to the fact the GPU has enough spare processing to read/write the larger data needed, making the experience of the user much more more smooth and the GPU more efficient.

Cache

The cache of the GPU shares the same system of the CPU. It has 3 main cache levels in which it operates in. These cache levels are the L1, L2, and L3 cache levels.

To elaborate, these are different aspects of a cache, which are used in different places and vary in size. Firstly is the L1 Cache. This Cache is typically the better performing out of the rest, but this is due to the fact that it is also the smallest. The L1 cache is always first one that is accessed by the system core.

The L2 cache is slower than L1 however that's due to the fact it is larger in capacity, so it is used to store the larger data. The L2 cache is the second cache where the system core will search if there was a cache miss in L2. Lastly is the L3 Cache, which is the largest in data storage but the slowest cache. This cache is shared between all cores and it is where the shared data between cores is stored.

Overall, having these different segments makes the whole process of finding the information needed a lot more efficient, and therefore quicker. Due to the fact there are 3 specialised cache variants to deal with different sized data that needs to be transferred or stored temporarily.

Address/Data bus

The same buses that are used for data travel within the CPU are also connected to the GPU. For instance, the data travels along the data path to the Monitor, where the information gathered by both the CPU and GPU are displayed for the user to see. Like the CPU, the GPU uses an Address/Data bus system.

With the address bus being measured by the amount of memory that the system can retrieve, more specifically specifying a physical address of a system. The reason being is that the computer processor needs to read/write from or to the memory.

While the data bus just simply carries the data, whether it is through read or write operations. A write operation will consist of the GPU writing the data on the data bus, while a read would have the memory controller get the data from a specific memory block in order to store it in the data bus.

Pixels

Within a GPU there is always the presence of 2D processors. Essentially, these CPUs process all the information they are given into calculations, so that they may appear on screen for us to see in the form of pixels. These 2D processors are there to render the pixels that are seen on the screen.

Which is a complicated and always active job, considering how the user is always interacting with the screen. For example, if the user is playing a 2D game or browsing through the internet, the 2D processor within the GPU will constantly be active on changing the pixels on screen and rendering it in order to give it a visual quality.

Overall, the 2D processor is responsible for the rendering of a graphic, which comes at the very end of a process. However, it is the rendering itself which determines what appears on screen, therefore is a vital component in the interaction of the PC.

Polygons

Additionally to 2D processors, 3D processors are also found within GPUs. Equal that to 2D processors, 3D graphics also have the ability to render objects on screen. However 3D rendering takes a lot more processing power, since it is rendering polygons instead of pixels.

To elaborate, more intense rendering is much needed for 3D processing due to the fact the calculations that are needed to work out how many objects and polygons will be displayed, is a lot more demanding on the computing than it is for calculating pixels for a 2D image. Usually, the rendering engine can consist of a dedicated hardware or software program, which generates a digital image in a "Frame buffer".

For example, in 3D-imaging, a wireframe is created with straight line then rasterized (filled with pixels essentially). Meaning that, when the processor then creates the colour, texture and lighting as it is stored within the memory. It then sends this data to the screen/monitor with a cable.

Overall, due to the intense calculation needed due to this complicated rendering, which is what is required for many modern games on the PC due to the vast amount of 3D models and graphics, a much more powerful dedicated GPU will be needed to run it smoothly. Hence why the most powerful GPUs on the market are more successful than the most powerful CPUs. Which may lead to a badly balanced computer in certain cases.

Graphics Memory

Graphics memory is basically the amount of information the graphics card can store. This includes the number of frames, the gamedata, graphics and other effects rendered in the game. Essentially, the more memory the graphics card has to store, the more information it can potentially process.

For example, if a GPU has the ability of containing more data than a majority of GPUs, it doesn't necessarily make it faster in performance, which is essentially how many frames per second (fps) the game runs. However, it gives it better potential in performance than other GPUs over a continuous time, due to the GPU being able to contain the build up of data.

Overall, graphics memory is as essential as it's performance, due to the fact that it needs to be able to store and use a lot of data in order to keep it's stability while the user is carrying out an intensive task, such as a video game. Basically, if a GPU with a high performance specification and very little graphics memory runs a game, it would perform well initially then start suffering performance issues over time.

Memory

RAM

RAM, or Random Access Memory, is the memory that the company can use on "random" occasions. Essentially, this is the memory that is kept available for the programs to use. The memory available is measured in gigabytes (GB) and the speed is measured in megahertz. This means that the more RAM a computer has, the faster it will operate and will be able to operate more software at the same time.

For example, the more graphically or overall hardware specification demanding games usually demand more gigabytes in RAM. Possibly due to the large texture files or large polygon count of a video game. Which has too much temporary data to be stored within 2 to 4GB (rarely over 8GB) of storage.

This memory is known as volatile memory, as it is lost when the system is powered off. Which is why games need to be saved, so they go to the storage which is the hard drive. Dissimilar to ROM (read only memory), which cannot be lost or altered. ROM does basic necessary functions such as booting up a computer system, as it contains the BIOS.

Overall, a hard drive typically is a storage device which uses a magnetic disk to keep the data. While RAM is a flash memory device, which keeps the data until the system is shut down. The reason this is useful for programs such as games, is that the RAM can contain all the temporary data which doesn't need to be stored, instead of the slower and more useful hard drive taking in the unnecessary data.

RDRAM

Also known as Rambus Dynamic Random Access Memory. This is a memory subsystem which promises to transfer up to 1.6 million bytes per second. It essentially works in a similar fashion to RAM, but also has the functionality of controlling the RAM itself and creating a memory subsystem in form of a bus, or a path, which connects the RAM to the microprocessor and other devices which use it.Essentially, RDRAMs existence is to transfer data at faster rates, basically as the direct links between systems.

However, RDRAM rapidly went obsolete with the launch of DDR 400, due to the fact it didn't keep up with the data transfer speed or the low latency rates. Overall, RDRAM is very rare to find within a modern computer system, purely due to the fact it has been replaced by a more efficient process which made it effectively obsolete.

SRAM

SRAM, or Static RAM, are usually made up of 4 to 6 transistors, plus wiring, and retains its data in its memory as long as the computer has sufficient power. Meaning that it does not need to be refreshed constantly, which makes this a much faster variant of RAM but also a more expensive one.

DRAM

DRAM, or Dynamic RAM, is memory which stores various bits of data in cells consisting of capacitors and transistors, millions of these will make up the RAM. With DRAM, the memory must be repeatedly refreshed or it will remove the data that was being contained. This process takes more time and slows down the memory. While slower, it is much cheaper than SRAM.

Display

Connection Type

Typically, there are a couple of connection types what are common within the modern world PCs. The most common connection types include HDMI (High-Definition Multimedia Interface), DisplayPort and VGA.

Firstly, HDMI is a unique connection type due to the fact it has the ability to carry both uncompressed video and uncompressed audio. Hence why it's become a widely used connection of choice for a majority of multimedia devices, as it's essentially a solution via one cable.

Other benefits that HDMI has include functions which include HDMI-CEC (HDMI Consumer Electronics Control), which essentially allows the user to control multiple devices with a single remote. For instance, a user could connect a soundbar to a television through a port which is HDMI-CEC compatible, giving the user the ability to control the television and the soundbar simultaneously through a single remote source.

Secondly, DisplayPort is a much more modern connection type, with the ability of carrying 3840 x 2160 (4K) resolution video at a 60Hz refresh rate, which is similar to 60 frames per second. The way this is done is through the 17.28Gbits/sec of bandwidth.

The benefits that DisplayPort has is the ability to output multiple displays through multi-Stream Transport (MST). Which can be done by connecting multiple compatible monitors over DisplayPort or by connecting a DisplayPort MST splitter to a single DisplayPort output on a PC.

However, there are limitations in using this. Since there is a bandwidth limit to DisplayPort, with the 1.2 configuration the user is only able to have two 1920x1080 monitors. While with the 1.3+ specification, the user can use two 3840 x 2160 displays.

Lastly, VGA (Video Graphics Array) is a very old connection type, making it the oldest out of the three. It was used with CRT monitors decades in the past. While VGA is technically output to 1920 x 1080p, it is an analog connection, which means that the resolution can be pushed higher through image degradation, usually caused by the conversion from analog to digital.

Overall, display connection types each have their own quality and purpose. With more modern connection types being able to support much higher transfer rates over multiple monitors, such as the DisplayPort. While older, more simpler connection types, such as VGA (and now HDMI) have the straightforward task of creating a 1920 x 1080p resolution display. It has to do with personal preference in terms of gaming.

Built-in

Built-in displays are relatively similar to external displays in the respects of it’s use, as they still allow a user to see the feedback from the GPU and the rest of the hardware into the software and operating system. However, unlike external displays, they are almost impossible to interchange screens. With PCs, the most common form of built in display is a laptop, although there are some computers that are built into the monitors.

Laptop screens are generally and directly attached to the rest of the machine, meaning that removing the screen would mean disassembling the laptop. Inturn, needing to find an exact replica in size and specification in order to change it. So this usually means that in order to upgrade the display, an unnecessary amount of cost would impact the user. This therefore shows that laptops tend to go “out of date” a lot quicker than desktop computers, as they don't have as much of an option to upgrade which desktop computers do.

Overall, it is debatable if built-in screens for PCs are in good use for gaming, despite their lack of portability. In other gaming platforms, built in displays can be seen within the handheld systems such as smartphones, as their screens are built in due to the fact it utilises a touchscreen feature to access the phone, which each new smartphone getting an upgrade each year.

External

An external display is essentially a device which allows the user to see what activity is going on within the software and operating system computer. For example, the most common example of an external display is a monitor, as it can be interchanged or removed completely from the rest of the system.

Another example could be a TV. While their main purpose was not necessarily for computers, but it is still an external display as it can be connected to a computer, usually by a HDMI cable, to allow it to act as a monitor, which has its own advantages.

For example, TV's are generally bigger than an ordinary monitor, not always however. Secondly, while using an external device such as a TV, it is possible to have more than one display available at the same time, assuming the computer has enough ports to support multiple external displays.

This can have a variety of uses, especially within the gaming genre. An example of this is that if a user wanted to play a game, but also was occupied with another task at the same time, such as watching a film or browsing YouTube. This it makes this possible and easier, not needing to switch and exit programs, as one screen can house one program, and the other can house the other. In addition to this, if a TV were to be the secondary external display, it would make it possible to play computer games on a bigger screen, sometimes with HD options depending on the device.

For example, high end monitors, such as the ASUS ROG PG278Q Black 27" Widescreen. Boasts an array of useful and debatably important aspects when it comes to gaming, all of which comes down to the monitor itself, not the PC. Aspects such as a 144Hz refresh rate, making faster frame rates and minimal motion blur, or the 2560x1440 resolution screen. Which allows for roughly 77% more on-screen awareness, and providing more detail than what is considered conventional, such as 1920 x 1080 screen. Essentially creating a bigger advantage to the player.

Sound

Sound-cards

All computers have a sound card included, it may either be integrated on the motherboard or a sound card in an expansion slot. Before their creation, PCs could only make small bit (or beeping) sounds.

The minimum that a sound card will have is four components, which are: The analog-to-digital converters (ADC), digital-to-analog converters (DAC), ISA/PCI interface to connection the sound to the motherboard and input/output connections for speakers and microphone. While some sound cards use a coder/decoder chip known as a CODEC that performs as a ADC and DAC in one.

Usually, when a user plays a sound into a microphone, an ADC will function by translating the analog waves into digital data by taking measurements of the sound waves at recurrent intervals. The speed and frequency of these measurements is called the sampling rate. Which is measured in kHz, with the more kHz, the more accurate reconstructed sound will be.

The DAC works the opposite way by reconstructing the original sound, with a high sampling rate the playback can be almost as identical to the original. However, to transport soundwaves through wires causes distortion, which reduces the sound quality. Which is more important for video games as people believe. Since sound is what makes a video game's experience genuine, it is vital to have the sound working in perfect order to function correctly.

Drivers

Drivers, otherwise known as hardware drivers, is essentially a group of data or files that enables one or multiple hardware devices to communicate with the computer's OS (operating system). Making them an integral part in computing, since a computer wouldn't be able to send or receive data sufficiently to hardware devices.

Drivers differ in quality and use depending on what the driver is for and which company designed the said driver. Typically, drivers for video cards and processors come from companies known for their high specification hardware, such as Intel and Nvidia. Therefore, the most common drivers for the vital hardware components may already be suitable for video games.

Built-in/External Speakers

Speakers are essentially a hardware device which is designed to generate sound from a computer, specifically from it's sound card. Typically, speakers can come in either built-into the device or have to be purchased and installed externally to the device.

Usually, built in speakers are found within portable devices such as laptops rather than desktop computers. Meaning that it would be extremely uncommon to find a gaming computer with inbuilt speakers, unless it we're a laptop. Usually, these internal speakers lack in quality and are mass produced only to have sound which may be considered "optimal" for the user, but not completely satisfactory.

Therefore, if the user was to get a better sound quality, they would need to purchase external speakers. External speakers are capable of being much higher quality due to the fact they are manufactured individually from other hardware. Due to the fact they are the center of design and manufacturing and have the ability of having much higher quality components and materials.

Game Format

File Size

The file size of the game is what determines how much data is stored within a single file. For example, the specified file could be a word document with the file size of 20kb or an executable file with a file size of 50GB.

Overall, file size is important to determine how much data a program has, due to the fact the PC might not be able to handle that much data. However, with the sophistication of modern computers, it is becoming less of a problem with file size interfering with performance. Unless the hard drive is full or almost full of data.

Physical

A physical copy a video game would usually come in optical disk. This is the name given to the disk format which both DVDs and CDs use. The reason for their popularity is due to their low cost in manufacturing, as well as the fact that they are impervious to many environmental threats including this like power surges, or magnetic disturbances.

They're also a lot more reliable and interactive that its predecessor, the magnetic tape. Optical disks can hold more data when compared to magnetic tape too, because of the greater accuracy with laser beams in comparison to the tiny magnetic heads used within the tape. This means the data can be compacted more. However, it is a problem to use optical disks for video games as they require a lot more data, and the general trend is the newer the game, the more GB it will take up.

Current gen games can take over 50GB of data to be installed before being ready to play, which is why blu-ray optical disks are becoming more and more popular, as well as digital download. Which would take a lot of optical discs to store, considering the fact a CD-ROM has 737 MB total capacity and a DVD-ROM has the capacity of 4.7 GB.

Download

Due to physical copies of video games slowly becoming obsolete due to the ever increasing file size of video games. Hence why games are also becoming digitized a lot more, not even requiring a disc but rather being downloaded straight from the internet.

Not only is this new form of game format caused by the limited capacity of optical disks. This new platform of gaming is also a result of new internet speeds and capabilities, as well as bigger hard drives to store the game’s full files on.

Overall, it is becoming more apparent that digitally downloading games is going to replace any means of game format. With the digital downloading becoming a much more superior variant to a physical copy.

Network

Wired and/or Wireless Network Adapters

A wired network adapter is an adapter, usually an ethernet port, which has some sort of cabling connected to a device which provides internet access, such as a router. With internet access almost instantly, without the need of a password. Usually, wired tends to have faster download/upload speed and latency when compared to wireless, however wireless technology has been improved to almost equal, if not better, than wired network technology.

A wireless network adapter is a device in which is responsible for searching and connecting to the various signals produced by a router, which is usually a Wifi signal. This Wifi signal provide internet access to the user connected, equal that to wired network adapters, if given correct access and rights to do so.

Download/Upload Performance

The download/upload performance of a network is essentially the read/write variant of transmission but over an open network of computers rather than the system itself. However, insteading of the system reading, it is downloading files to the storage (hard drive), and instead of writing, it is uploading data from the storage (hard drive).

Latency

Latency is essentially the delay in a network between the host system and the outcome, whether it is a program, a web page or a player on an online video game. Usually, more latency isn't a good thing because it delays the process of the system, which can cause errors and frustration for the user. Essentially, it's the short amount of time it takes for an action to occur, usually over the internet, which is undesirable at an extreme delay.

Power Supply

Internal/External

Most PCs have a PSU (power supply unit) within them. The Power Supply Unit basically acts like the battery, or the main powerhouse, for the computer. It is becoming more apparent that modern PSUs have some type of UPS (Uninterruptible Power Supply) battery backup.

Essnetially, the UPS battery is designed to protect the computer's data and storage from harmful events such as power surges or cuts. This is to prevent these unexpected losses of power, which have colossal potential to be harmful to data. Especially data which has not been stored properly, sometimes corrupting files.

Hence why UPS batteries act as a battery backup. Helping to have a more reliable power source that goes into the computer, instead of facing the risks of data loss. However, while modern, these types of batteries are only really found within expensive computers.

This includes servers, and more specifically game servers. By having this countermeasure and extra protection, it will reduce the risk of disrupting players if the online game servers go down, and will therefore make for a more enjoyable experience.

Overall, PSUs only purpose is to provide sufficient power to the computer, with some PSU having a degree of protection with an extra cost. This is to prevent any loss or corruption of data.

Console

HCI

Interaction Models

Although it is important for a device to be able to interact with the computer well, it must also interact with the human sufficiently, hence Human-Computer Interaction. Meaning that it doesn't matter how technologically advanced a HCI device is, it still needs to be usable by a human at satisfactory levels.

Ergonomics

Hence the importance of ergonomic design. Ergonomic design refers to how a device is made to be held or used, and whether or not it is easy to use. This can be decided simply by using it, and seeing how easy it is to either hold it, or interact with it to achieve what you need to do on the device with ease.

The ergonomics of a console controller is arguably what makes them so popular. Through the design of controllers, there is a strong correlation of how they are held and used, most notably in the range of PlayStation controllers.

They are designed to be held by the player using both hands, and for the person's fingers to be able to use both analogue sticks, the direction pad, and the buttons. As well as the triggers and buttons on top. With all of these different buttons and controls required to play most games, as well as function the menus of the Playstation itself, it needs to have a comfortable and well thought out Ergonomic design.

Button Configurations

As briefly mentioned above, the location of the buttons play a key part in what makes a HCI comfortable and easy to use. This can be seen when looking at the conventional game control once again.

There are four buttons in a plus placement located on the right, and a directional pad which is also shaped like a plus on the other side. While the PlayStation controller is unique to the competitor's controller, the principle still remains the same.

This is that by making the controllers button configuration symmetrical, as can be seen from the PS4 controller, it makes the position of both the buttons and d-pad a much better configuration, as it provides a much easier way to control what's going on in the games.

This is because not only are they placed in the most optimal position for the finger placement, but also because the 4 buttons and the directional pad are some od the most used controls, and so by putting these in the most comfortable place, it means the player will be in this position more than anything else.

Inputs & Control Mechanisms

Wired

Wired control mechanisms is something that Playstation hasn't incorporated within their systems ever since their Playstation 2. This may be due to the fact that wireless has a much more extensive range than wired. Also due to the fact wired connections cause a hazard for individuals, especially in common areas where consoles are found such as living rooms and bedrooms.

Wireless

Wireless control mechanism are very popular amongst the Playstation system, especially with the introduction of the Dualshock 4 controller which uses bluetooth. Due to this, the PlayStation 4 is most notable within the market to use Bluetooth, as well as their headsets.

The reason that Dualshock 4 is using bluetooth, is due to the fact that it reduces the amount of wires required for technology, allowing for a less hazardous space and for more freedom of movement. This is done by using a special radio frequency to transmit data, the downside to this however is the lack of range, meaning there will always need to be a main transmitter nearby.

Therefore, the reason that Dualshock 4 controllers are wireless is due to their extended range, extra mobility and less hazardous space. Essentially, making it more comfortable for home use rather than the HCI of the controller itself.

Touch-Screen

Touch-screen technology is an aspect that isn't currently present with Sony's PlayStation 4 system. Due to the fact it is completely trivial for console use and it is therefore seen as unnecessary.

CPU

Type

The CPU of the PlayStation 4 has 8 cores. Which is considered good by modern standards, but cannot be determined on what the performance of the processor is. Since the cored only define the multitasking ability of the PlayStation 4. Which is necessary due to the fact it is constantly recording video game footage in the background. Also, there are users who stream via the PlayStation platform, making it more demanding on the multitasking aspect.

Essentially, the 8 core CPU is a very good standard for a console, especially for the most produced and sold console in the world. Especially due to the fact that it has the ability of constantly recording gameplay without the console crashing or overheating.

Speed

The AMD Jaguar CPU in which the PlayStation 4 has has a clock speed of 1.6 GHz, with a 2.75 maximum frequency. Which isn't considered good by modern standards of video gaming, but for console games, it is possible for it to be optimised around. Essentially, the clock speed isn't good when compared to high end machine such as PCs, but for console standards it would be considered "good enough" for video game optimisation.

Cache

The cache on the PlayStation 4 CPU is a Shared L2 Cache, which is 2 x 2 MB. Which is the balance between data transfer speeds and data capacity, which is good for a gaming console. Since the user wouldn't want the data to be too slow or too little at a time.

As mentioned previously about cache, there are 3 different types of Cache, such as L1, L2 and L3. With L1 being the smallest in capacity and L3 being the largest in capacity but the slowest. Hence, with the use of L2, it is a perfectly balanced system for a console, which is essentially a medium end spec computing device.

Address/Data bus

The Address/Data bus of the Playstation 4 CPu is x86-64, meaning that it is a 64 bit bus, which basically means 64 bit computing. This gives the PlayStation 4 the ability of addressing 16EB of address memory space.

However, this doesn't guarantee that all the games on the Playstation 4 run at 64 bits, since x86-64 Address/Data bus both supports 32 bit and is 64 bit by default. Although, this gives the Playstation 4 the ability to run games at a much smoother performance if the game was to support the 64 bit architecture.

GPU

Type

The Playstation 4 contains a GPU which is “essentially” the AMD Radeon HD 7850, which has an interface of a PCI Express x16. With claims that the best AMD APU that is currently available on the market, which is the Richland A10-6800K, has roughly a third of the processing power of the GPU found in the PlayStation 4.

Essentially, the Radeon HD 7850 is considered a medium end graphics processor, but in comparison the the previous generation of consoles, it is a colossal leap in hardware technology. Meaning that the mass produced console is getting closer to photorealistic graphical capabilities, which is a good sign for the evolution of technology and gaming as a whole.

Speed

The GPU of the PlayStation 4 is a graphics card manufactured by AMD, which was launched in 2013. It is claimed that it is based on the “Liverpool” graphics processor in a specific variant, which supports DirectX 11.1.

The PlayStation 4 uses a custom AMD Radeon GPU that runs at 2.13 GHz, with a 1,843 GFLOPS (floating points operations per second). Which is essentially the vertices/triangles per second and the pixel fill rate. These speeds measure how fast it can build a wireframe and how many pixels are processed per second.

As mentioned previously, the overall performance of the graphics card is determined by its frame rate. Overall, in terms of video games, they require this process every second and most games are expected to run at 60 frames per second (fps).

Cache

The cache of the PlayStation 4’s GPU is using 512KB of L2 cache. Which, as I have mentioned previously, the L2 cache is a well balanced cache between data transfer and performance. Even though 512KB may not seem like enough, it is good enough for the PlayStation 4’s heavily optimised system and video games in order for it to run the games smoothly at 60 frames per second, while still maintaining the background applications.

Address/Data bus

The Playstation 4’s GPU features a 256 bit bus width. Which is outstanding, especially for a console which is mass produces and widely sold around the world. This proves that it is good for higher demanding games at an optimal level, due to the fact it can transfer a large amount of address memory space. Increasing the efficiency of the console when dealing with heavy applications such as video games.

Pixels & Polygons

The Playstation 4 also has an impressive array of raw performance. Featuring 1,152 shading units, 72 texture mapping units, 32 render output processors, 18 computer units, a pixel rate of 25.6 GPixel/s, Texture rate of 57.6 GTexel/s and an impressive 1,152 shading units.

What this means is that it has much higher capability of real time rendering when compared to it's older counterpart. Meaning that the evolution into the next generation was a significant leap from low resolution 3D polygons. To highly detailed 4K resolution textured 3D models with advanced shading and anti aliasing. Essentially, this type of rendering gives the PS4 that "crisp" look in graphics when compared to the older generation of consoles.

Graphics Memory

The PlayStation 4 features the graphics card RAM of GDDR5 SDRAM with 2GB of memory, running at 153.6 GB/s, with the memory clock speed at 4.8 GHz. This is also a significant improvement to the predecessor which had 256MB of GDDR3 RAM and a clock speed of 1.3 GHz.

Essentially making the PS4 the much more superior console when compared to older generations. Which again, is great for the evolution of video games because it solely means that technology heading the path of being improved over time, especially with the newer PS4 Pro.

Memory

RAM

Not only is the GPU on the PlayStation 4 impressive, the RAM is also a colossal improvement over the older variant of the console. The Playstation 4 features a GDDR5 8GB RAM With a frequency of 5500 Mhz, which gives it a much higher ability to multitask programs, alongside the powerful processor that it already has installed.

Furthermore, the Playstation has a System Memory bus of 256 bit. Which, as I explained earlier, gives the PlayStation the ability to handle more demanding applications better, and therefore increasing its performance and multitasking abilities. Especially with the system memory bandwidth of 176.0 GB/s, which is very impressive for a console in the price range that it currently is at.

RDRAM

RDRAM isn't used on the PlayStation 4 system since it was considered obsolete by the mid-2000s.

SRAM

No evidence of SRAM found within PlayStation 4 system.

DRAM

The Playstation 4 is believed to have 1GB DDR3 DRAM, which is considered average for a console of this generation, but yet again another improvement to the hardware specifications. Giving the Playstation the ability to transfer data at a faster rate than the previous generation, which would in turn decrease loading times, especially for more demanding applications.

Display

Connection Type

The Playstation 4 only has a HDMI display connection type available. Which is an outstanding connection type due to its universal and wide use on the modern day, especially due to the fact that most TVs have HDMI ports. Therefore, it isn't much of a limitation having only HDMI on the Playstation 4 with modern equipment.

However, with the Playstation only being limited to HDMI means that it is no longer compatible with older connection types such as VGA, or newer connection types such as DisplayPort. Although, VGA has slowly been replaced for HDMI over the years due to HDMI's superior cabling and architecture. DisplayPort is proven to be much more beneficial than HDMI, almost making HDMi obsolete which is possible in the near future.

Overall, HDMI is a good choice for gaming given the current circumstances, but it is impossible to have backwards compatibility without the use of a HDMI/VGA Adapter, giving extra cost and unnecessary energy. Furthermore, the Playstation 4 in its current specification isn't very "future-proof" due to the fact it doesn't have a DisplayPort connection.

Built-in

The Playstation 4 platform does not feature a built-in monitor, due to the fact there is not need to have a screen on a console device built for TV use.

External Screen

The external screen in which the Playstation 4 relies on is a home Television or a monitor due to the available and universal connection type of a HDMI cable. Which, as explained previously, is beneficial with the current circumstances on technology.

Sound

Sound-cards

Digital signal processing SIP block (Identical to TrueAudio). Which would be considered a mediocre sound card, however for the price range of the PlayStation 4, it is very good considering the quality to price ratio. Which means the player will get a more enhanced sound experience as well as a graphical and performance one.

Drivers

The Playstation 4 uses AMD next-generation Radeon based graphics engine. Which is an engine that is fully compatible with the components inside, because it needs to be otherwise the console wouldn't function properly. Additionally, these drivers we're designed for video games, therefore having a lot of optimisation due to the drivers.

Built-in/External Speakers

No evidence of built in speakers on PlayStation 4 since it relies on the use of a TV or external speaker. However, the Dualshock 4 controller has a small internal speaker facing the player which has audio feedback depending on the game they are playing.

Game Format

File Size

The Playstation 4 has a storage space of 500GB, which is a good amount of space to have a couple of video games on. However, as the file size of video games increase, the less game files are able to be saved within the PS4. Especially games that can have their file size larger than 50GB. Which would mean that a maximum of only 10 games could be installed within a Playstation 4 without any additional save files or downloads.

Physical

The blu ray disk that the Playstation 4 uses has a capacity of 50GB, which is becoming a more average game size with new releases, due to the amount of files required in order to create a graphically competitive game in the modern market. This in turn means that blu ray will soon become obsolete, the same way DVD and CD became obsolete due to their capacity limit.

Download

However, the PlayStation 4 has the ability of downloading games standalone without a physical copy. Meaning that, even if the blueray does go obsolete, the Playstation will still be able to download video games off the digital market without the need of a physical copy.

Network

Wired and/or Wireless Network Adapters

The PlayStation 4 has Gigabit Ethernet wired adapter, which is a standard ethernet port but is proven to be very effective and useful even to modern standards. This is simply due to the future proof architecture of ethernet cable design.

However, the Playstation 4 features a 802.11 b/g/n Wifi wireless adapter, which is considered somewhat poor by modern standards. With the introduction of fibre enabled internet connections with 5Ghz, the internet has been much faster to browse, but that wouldn't be apparent with the Playstation 4 since it doesn't support the modern bandwidth of 5GHz.

Meaning that it wouldn't be able to fully utilise the potential of fibre enabled Wifi, making the internet performance loose a lot of potential due to the Wifi connection using an almost outdated antenna.

Download/Upload Performance

As mentioned above, the download/upload performance of the Playstation 4 is rather limited with Wifi due to the wireless connection only featuring a 2.4Ghz connection, lacking a 5Ghz connection which lower in range but is considerably faster than 2.4Ghz.

Latency

However, even with 2.4Ghz only wifi connection, the latency of the Wifi will have little to no effect. Meaning that playing online wouldn't be much of a different experience, unless the Playstation 4 starts automatically downloading updates or the player wants to stream the game they are currently playing.

Power Supply

Internal/External

The power supply of the Playstation 4 is an internal battery, which works similar to the PC's power supply unit but at a much smaller scale. However, internet batteries within a console is a lot more disadvantageous than most would like to believe.

For example, if a player was on the Playstation 4 system extensively over a period of time, the heat generated from the internal power supply would heat up the console, increasing the chances of the console overheating or damaging small but fragile components. Which would reduce the lifespan of the console.

Furthermore, since the power supply is internal, it is taking up unnecessary space as well as creating excess heat. Instead of the battery being internet, another more useful component could of been implemented. For instance, an improved wifi network connector.

Smartphone / Tablet

HCI

Interaction Models

The iPhone's main interaction model is the touchscreen feature, which gives it an extremely user friendly interface due to the fact a user could use their natural touch to navigate the entire operating system of the handheld device.

Ergonomics

Even though smartphones, including the iPhone 6s, are very simple in design, proving to be sufficiently ergonomic to be desirable in the hands of a user. The use of this interaction model isn't great with video games, especially specific games where you control the character's movement.

The reason being is that it's not as ergonomic as a controller is, or even a mouse and keyboard. The touchscreen cannot be adapted to play the same popular games. Due to the fact it is limited in customisation, even though it's extremely versatile to the human touch.

Button Configurations

The iPhone only features a small array of buttons which can be used by the user, due to the touch screen being the biggest interaction model of the iPhone. Those buttons include: the home button, the volume buttons and the sleep/on/off button.

Meaning that the iPhone is very limited in the versatility of types of interaction, while it does compensate for the versatility of the single type of interaction that is touchscreen.

Inputs & Control Mechanisms

Wired

The iPhone does not feature any form of wired control mechanism.

Wireless

The iPhone does not feature any form of wireless control mechanism.

Touch-Screen

Touchscreen technology is a very dominant control mechanism within the smartphone world in the modern market. In fact, it is the only mechanism which controls the operating system of the device. Due to its convenience and simple to learn/use method of HCI.

However, somewhat contrary to what has been said previously about user friendliness, touchscreen isn't very user friendly with video games. Hence why gaming PCs and consoles do not feature a touchscreen, unless it is a portable console. While portable console still have some form of analog sticks and buttons.

CPU

Type

The CPU of the iPhone 6s has the Apple A9, which is a 64-bit ARM-based CPU which is a system-on-chip, or SoC. Which is considered to be a very advanced CPU, due to the fact it can

carry out both the jobs of a CPU and a GPU, due to the fact it has a GPU chip built internally to the CPU.

Speed

The Apple A9 is a Dual-core 1.84 GHz Twister. Which is extremely quick for a smartphone processor, especially from the year 2015. This proves that the iPhone is a perfect handheld device for when a user wants to experience a portable device with powerful enough hardware to run even 3D games with effects.

Cache

The Apple A9 features an L1 cache, which has a 64 KB instruction and 64 KB data transfer per core; L2 cache which has 3MB shared data and an L3 cache which has 4MB data shared.

Address/Data bus

The A9 chip is built on 64-bit technology. Meaning that the processor is significantly more capable than standard 32-bit processors, due to the fact it is handling a lot more data at once. Furthermore, the A9 64-bit processor is capable of storing more computational values, including memory addresses. Meaning that it is able to access over four billion times as much physical memory than a standard 32-bit processor.

GPU

Type

The iPhone 6s uses PowerVR GT7600 (Six Core Graphics), which is a behemoth of a GPU for a handheld device as small as the iPhone. When the GPU was released in September 2015, it was labelled the “fastest GPU for smartphones” and still rivals the current GPUs found in tablets.

Speed/Cache/Address/Data bus/Pixels/Polygons/Graphics Memory

However, the specifications of the PowerVR GT7600 in the Apple A9 is not officially confirmed by Apple in any source, therefore the information is either confidential or possibly false.

Meaning that the GPU Base Speed, Cache Data, Address/Data bus, Pixel rendering, Polygon rendering and graphics memory of the Apple A9 is still confidential to the Apple company, therefore cannot be stated.

Memory

RAM

The iPhone 6s has 2GB of RAM. Which is the standard amount of ram on a smartphone in modern technology. However, Apple achieves comparable performance to its competitors with

less RAM due to the fundamental differences in how the iOS and Android platforms handle memory management.

For instance, Android relies upon a method called “garbage collection”, while iOS takes a “reference counting” approach. It is still disputed which method is better, however there seems to be a generally accepted opinion that garbage collection requires more memory to avoid performance issues,

RDRAM

The iPhone specification has no evidence of RDRAM

SRAM

The iPhone specification has no evidence of SRAM

DRAM

The iPhone specification has no evidence of DRAM

Display

Connection Type

The connection type on the display isn’t necessary on an iPhone since it is an internally built screen.

Built-in

Apple’s iPhone has a built-in touchscreen due to the nature of the smartphone design. This is due to the sophistication of the device and its human computer interaction capabilities of the touch screen ability the smartphone has.

External Screen

Due to the fact that the iPhone 6 doesn’t have a display connection type, it isn’t possible to have the iPhone connected to an external screen without the assistance of bluetooth or Wifi.

Sound

Sound-cards

The iPhone has a three main audio output devices available for the user to interact with or experience. These audio output devices include a 3.5mm jack, a Loudspeaker with 16-bit / 44.1kHz audio and a earphone with active noise cancellation via dedicated mic

The 3.5mm headphone/earphone jack is uses for the 3.5mm audio connection, which is the most commonly sold audio jack commercially. This is due to the universal yet good quality application of the 3.5mm audio jack and it's wide potential.

The loudspeaker is a factory installed speakers on the iPhone which Apple only refers to as a "Built-in speaker". Even though the sound audibility is sometimes too little to hear for certain people, the quality of the sound seems to be of good quality build.

Drivers

The Apple iPhone doesn't give any access to drivers or the source code, due to the fact it's entirely confidential to the Apple company.

Built-in/External Speakers

The iPhone features pre-built speakers what have been pre-built into the phone during production. Typically, the quality of these speakers are considered to be "good enough" quality, but given the size of the speakers they tend to be very high quality in comparison to others with a similar size.

Game Format

File Size

iPhone internal memory can range from 16, to 32, to 64 to 128GB of storage space. This is colossal storage space for a handheld device, especially with 128GB of storage. Which gives the iPhone the ability to download a lot more applications, especially applications larger in file size which was a challenge to obtain previously.

Physical

It is impossible for the iPhone to physically install a video game, therefore it doesn't feature the physical game format.

Download

Digitally downloading games on the smartphone is the only method in which games, and any other application, can be installed within the device. The way in which games are downloaded are through the App Store, usually found in the home menu.

Network

Wired and/or Wireless Network Adapters

iPhone only function through the means of a Wireless Network Adapter, due to the fact there aren't any available ports which give it access to wired networks.

Download/Upload Performance & Latency

The iPhone and it's network connection performance heavily depends on what type of router or wireless access point it is connected to, due to the fact that the iPhone's network driver and adapter is at a high standard, especially when compared to other smartphones from smaller competitors.

Power Supply

Internal/External

As rechargeable batteries are very common within smartphone and other mobile devices. It is also evident that the external source that is the charger is also common, as it requires to charge the battery itself.

This charger cable is a form of external source for power supply, as it is the aspect which converts electricity from a main power outlet, into electricity which can be stored within the Lithium-Ion battery. The use of such a cable is very useful as it is often portable, meaning that, even while traveling, the battery can still be charged.

This is another aspect to which it is useful, as most rechargeable battery cables use a plug, making it easier to find an eligible place to charge.

Most, if not all, devices that are ion charge can still be used. Meaning that if a portable device is no longer being used to walk around with, it can be turned into a static device with unlimited battery life. The most common scenario for this would probably be a laptop, as people often find themselves using it in the same room or place all the time.

However, excessive charging of the battery will eventually cause the battery to decay. Which would cause a vicious cycle of it needing to remain plugged in to keep it on for long periods of time.

This is prominent in people who are using technology everyday, they tend to find themselves playing for hours at a time. While it is often easiest to keep the device on charge the whole duration, so the user doesn't need to keep plugging and unplugging the device. Which can be the major disadvantage of a rechargeable battery.

Software Technologies for Configured Platforms & Devices

PC

Operating System

Typically, the Windows 10 OS isn't necessarily platform specific. However, it is usually found with the PC platform. The Operating System is also open to the user due to the wide accessibility to the operating system via the common platforms.

The Windows 10 Operating System has the purpose equal that to other operating systems currently available, which is to have the user interact with the hardware and software found within the machine, in order to carry out any purpose needed with the platform.

Due to the ability of an operating system giving users the ability of accessing the hardware and software of a platform/device. Giving the opportunity to create programs within the operating system for productive, such as an employee, or for recreational purposes, such as video games.

Typically, modern video games have a wide variety in visual development and capability. However, they are limited to the current hardware capabilities the devices containing the operating system currently have available.

Drivers

It is most that likely the drivers found within the Windows 10 OS are highly specific to the type of platform, even for the type of OS (i.e. Windows) and version of said OS (i.e. Windows 10). Usually, drivers are completely open for access to the user in the Windows 10 OS. Even giving them a wide variety of different drivers each having their own specific quality and use.

Sound

Drivers take information from the hardware input and output, which enables them to be understood by software which requires it. This is completed by translating the information into a language the software can understand.

This is essentially what allows game consoles to play and understand games specific to their console, as each console has a different driver with different translations. To elaborate, each device has their own specialised command line which only the driver knows, and the driver

accepts generic commands from a program and then translates them into specialised command for that specific device to understand.

Essentially, a sound card driver is exactly as described above, only it specializes in telling software exactly how to translate data into audio signals. This is so the card can then output to a set of speakers. Obviously this is a must when it comes to video games, as without a sound card driver, no audio would be able to be heard as the audio is not being processes.

Graphics

The driver specified for the graphics card essentially works in the same way, but instead of focusing on audio, it focuses on translating the data for video. This includes such aspects as lighting within a video game, or textures. Typically, the better the drivers, the faster the data will translate when needing to load.

This gives players more of an advantage if their drivers are better, as it means the game will load faster, allowing them to see the program much more clearly. This is why drivers are important to a PC, especially for gaming.

Overall, without the driver, there would be no video. This would most likely crash the game, as the computer software won't be able to keep up with commands it is being told without a translator.

Network

Also known as NIC for short, this is a form of driver which uses the internet in order to send and receive data. Essentially, it uses a connection such as WiFi or an Ethernet connection. With connection to the Wifi or ethernet signal, the NIC itself allows a device to connect to the internet.

Furthermore, the NIC allows the updates of the system’s software, download new games and even stream apps. As such, the newer generation of consoles like the Xbox One, PS4 and Wii U all depend on this driver.

Once again the importance of having a good driver is vital, as players gain a much better tactical advantage if their connection is strong and aren’t experiencing performance or latency issues. It also means downloading applications is a lot more time efficient, so they may get more applications at a much faster rate. Usually, most NICs feature a speed rating of either 11, 54 or 100mbps.

Overall, Sound card & graphic card drivers are essential software tools which command the operating system what to do with the hardware. Hence why updating drivers can often increase performance of games and reduce bugs and graphical issues. A NIC is a network driver that allows a system to connect to a network.

Installed Game Application Software

Video Game development varies with cross platform capabilities, which essentially means that it is the same game on different platforms. However, due to the significant OS and hardware difference of PC to other platforms, video games on the PC platform are completely platform specific.

Language

The most common language used for Game Application Software on is either C, C++, Java and even Python. However, these are the most common of video game development languages, there are significant amount of programming languages in existence, most of which are completely unique from each other and specialise in their own specific methods.

Object Oriented

All of the previously mentioned languages are some form of object oriented programming. This is a language where there are segments of code that is written as a list of objects. Which allows the grouped bits of code to be used in specific activities, depending on their intended purpose.

This type of coding fits perfectly with the games and mobile phone apps as it is very efficient in processing power. An example object-oriented programming could be the use of animals within a game, using a base "Animal" class, with "Cow" that have a different "EatGrass" method. By grouping codes into bases like this, it allows easy access to full objects required within a game.

Game Engine Scripting

Scripting is a program that is written run environments that are interpreted, which is then automatically turned into a task. The most common things that are automated through scripts are internet pages. This is because webpages use aspects such adverts, graphic displays, and menus, all of with run on scripting to be interactive.

An example of scripting can be related back to Java, as this can be used to develop websites. Scripting languages include QuackC, Lingo, Unrealscript, and Maya. All of these aside from Lingo are gaming engines, which allow the user to create models and assets for games when implemented into the Unreal Engine.

Graphical API

Firstly what is an API? Well short for Application Programming interface, this is a set of protocols and tools for building software applications. It expresses a software component, relative to inputs and outputs, as well as underlying types and operations.

It can also be referred to as a web tool. The point is that software companies may release its API to the public in order to get other software developers to design products using their service. An example of this could be seen when looking at Amazon.com.

They released their API so the website developers could access Amazon's product information a lot more easily, and allows third party websites to post direct links to amazon products with updated prices. It is essentially a software to software interface, as opposed to a user interface, as applications may talk to each other without the users knowledge.

Now specifically, a Graphic API is the software that renders the video you see in the screen. For example the creation of DirectX started off as a built in API which still is imbedded with Windows OS, and provides low level functions which give access to hardware. When the Xbox was first named, the X within the name came from DirectX, in order to indicate that the console was based on this technology.

DirectX is actually various APIs, all starting with the word Direct, and so X as a shorthand. The most recent being DirectX 11, which is seen as revolutionary by some and controversial by others. Which is probably due to Microsoft's heavy partnership with Nvidia and Intel

Sound API

Pretty much the same as a graphic API, however instead of focusing on visuals, it focuses on audios instead. There are two main sound API which i will be talking about, and this is OpenAL, and Java Sound API.

This API is designed to help users write programs, and is a low level API which controls the inputs and outputs of sound, it has an excellent degree of control over audio capability. It is a pretty straight forward API when compared to the others, as it follows the already mentioned conventions.

Console

Operating System

The operating system for the Playstation 4 platform is Orbis OS, which is a heavily modified version of the FreeBSD operating system, known for being similar to the Linux OS. However, not much is known about this operating system due to the fact that Sony is very confidential about their company data, especially the programmes used for their console systems.

Drivers

Sound/Graphics/Network

Due to Sony's confidentiality on the subject of software drivers, the sound, graphic and network drivers cannot be listed since there is no clear evidence or confirmation from Sony about these specific software systems. However, all that is known is that system updates do have an effect on the said drivers, as they sometimes require upgrades.

Installed Game Application Software

Language

On consoles, including the PS4, the most common programming language used is either C# or C++, with C++ being the mostly used one out of the two. Other languages are also practised, such as Lua, Python, JavaScript, Tcl and Perl.

Object Oriented

The programming languages listed are Object Oriented, which is fantastic for video game design. As mentioned previously, object oriented coding fits perfectly with the games and mobile phone apps alike as it is very efficient in processing power.

Game Engine Scripting

The Unity Engine is the most common engine with consoles due to their ability and easy to make video games optimised when compared to other engines. However, there are more games being announced with the revolutionary Unreal Engine 4, so the Unity Engine may be replaced by a more superior Unreal Engine 4.

Graphical API

Although unconfirmed, it is believed that the Playstation 4 features two different graphics APIs. Which are: GNM, which is a low level API and GNMX, which is a high level API. Apparently, the high level API, GNMX, is what most developers commonly start with, which they use to "wrap" around the GNM low level API.

Additionally, it is said that the PlayStation's API will be able to "hold its own" against DirectX 12. However, this is all unconfirmed by Sony so their API systems and their capabilities are speculative to the mass population.

Sound API

The sound API used on the Playstation 4 operating system is called Audio DSP (Digital Signal Processor), which is seen to be heavily based on AMD's TrueAudio API. Sony utilizing

TrueAudio on the Playstation 4 platform potentially means that there will be at least some level of audio engine portability between the Playstation 4 and the current generation AMD products.

Smartphone / Tablet

Operating System

The operating system of the Apple iPhone is the iOS operating system. Not much in known about the iOS operating system since the source code is closed to the public. However, what is known is that it is Apple's own design and creation which was made to run on their iPhones, iPads and iPod touch devices.

The operating system, being completely reliable on touch screen, includes a multi-touch interface where the user can make simple gestures to operate different functions within the device.

Drivers

Sound/Graphics/Network

Apple has provided no evidence in the sound, graphics and network drivers that are used within the iOS platform. Due to this, the information is most likely classified to within the company and cannot be listed. Although, it is known that system updates do upgrade the drivers which have the responsibility of these three aspects.

Installed Game Application Software

Language

Objective-C, otherwise known as Swift, is the language used within the iOS operating system, which is a programming language created by Apple themselves. Though it is suspected that some C++ is embedded within the iOS platform.

Object Oriented

Apple, the creators of Swift, confirm that the programming language is object-oriented in nature. Essentially, Swift is a "powerful" and "intuitive" programming language designed by Apple, which is meant to be "interactive" and "fun". Apple claims that the syntax used for Swift is "concise" yet "expressive".

However, Swift is a programming language restricted to only Apple made products and operating systems. Which means that it's only available in macOS, iOS, watchOS and tvOS. Although, it is claimed that Swift code is designed to be safe and produces software that runs "lightning fast".

Game Engine Scripting

Sparrow is an extremely common engine, which is also open source, meaning that the public has access to the source code for free. Sparrow is apparently a pure Objective-C library that was built “from ground up” specifically for the iPhone, iPad and iPod Touch. While there are multiple engines for mobile gaming, such as unity and unreal engine, Sparrow seems to be the most commonly and reliable out of the rest for iPhone applications.

Graphical API

The graphical API for smartphones, more specifically the iPhone 6s, is the OpenGL ES 3.0. OpenGL, or Open graphics Library, is essentially a programming interface tool which is used to render both 2d and 3d vector graphics. Usually OpenGL is a cross-language and cross-platform application, meaning that it’s compatible with many devices with many programming paradigms and languages.

While Apple’s GPUs is suspected to support Metal, alongside with OpenGL. Metal, which is an Apple created 3D graphics render application, is designed to perform data-parallel computations using the GPU. Essentially, this application was created to gain low-level and low-overhead access to the graphics processing unit, enabling the user to maximise the graphics and compute the potential of the application.

Sound API

Apple claims that “both iOS and OS X come with a wide array of audio APIs”, which can range either from low to high level. It is claimed that the AVFoundation is available both on iOS and OS X, which is their “modern” media framework which contains multiple APIS for “different purposes” and on “different levels of abstraction”. With the code consisting of modern and low level C language for the APIs.

AVAudioSession is the API specific to iOS, meaning that it’s only found within iPhones and iPads. Which is apparently an API which coordinates the audio playback, depending on the apps the user is on.

For instance, audio stops when a user gets a call, or the music playback stops when the user watches a video such as a movie. They claim that it is needed to make sure an app behaves correctly in response to these specific events only found on the iOS.Furthermore, on iOS, Core MIDI and COreAudioKit are used in the development of a mobile app, in order to behave as a MIDI instrument.

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